

# **FANUC Robot series**

**R-30*i*B Plus/R-30*i*B Mate Plus  
CONTROLLER**

**CC-Link Interface (Slave)**

## **OPERATOR'S MANUAL**

MAROCUCCLK0817E REV. A

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**B-83974EN/01**

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FANUC conducts courses on its systems and products on a regularly scheduled basis at the company's world headquarters in Rochester Hills, Michigan. For additional information contact

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For customer assistance, including Technical Support, Service, Parts & Part Repair, and Marketing Requests, contact the Customer Resource Center, 24 hours a day, at 1-800-47-ROBOT (1-800-477-6268). International customers should call 011-1-248-377-7159.

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### **FANUC America Corporation Patent List**

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## Conventions

### **WARNING**

Information appearing under the "WARNING" caption concerns the protection of personnel. It is boxed and bolded to set it apart from the surrounding text.

### **CAUTION**

Information appearing under the "CAUTION" caption concerns the protection of equipment, software, and data. It is boxed and bolded to set it apart from the surrounding text.

**Note** Information appearing next to NOTE concerns related information or useful hints.

- **Original Instructions**

Thank you very much for purchasing FANUC Robot.

Before using the Robot, be sure to read the "FANUC Robot SAFETY HANDBOOK (B-80687EN)" and understand the content.

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In this manual, we endeavor to include all pertinent matters. There are, however, a very large number of operations that must not or cannot be performed, and if the manual contained them all, it would be enormous in volume. It is, therefore, requested to assume that any operations that are not explicitly described as being possible are "not possible".

# Safety

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FANUC America Corporation is not and does not represent itself as an expert in safety systems, safety equipment, or the specific safety aspects of your company and/or its work force. It is the responsibility of the owner, employer, or user to take all necessary steps to guarantee the safety of all personnel in the workplace.

The appropriate level of safety for your application and installation can be best determined by safety system professionals. FANUC America Corporation therefore, recommends that each customer consult with such professionals in order to provide a workplace that allows for the safe application, use, and operation of FANUC America Corporation systems.

According to the industry standard ANSI/RIA R15-06, the owner or user is advised to consult the standards to ensure compliance with its requests for Robotics System design, usability, operation, maintenance, and service. Additionally, as the owner, employer, or user of a robotic system, it is your responsibility to arrange for the training of the operator of a robot system to recognize and respond to known hazards associated with your robotic system and to be aware of the recommended operating procedures for your particular application and robot installation.

Ensure that the robot being used is appropriate for the application. Robots used in classified (hazardous) locations must be certified for this use.

FANUC America Corporation therefore, recommends that all personnel who intend to operate, program, repair, or otherwise use the robotics system be trained in an approved FANUC America Corporation training course and become familiar with the proper operation of the system. Persons responsible for programming the system—including the design, implementation, and debugging of application programs—must be familiar with the recommended programming procedures for your application and robot installation.

The following guidelines are provided to emphasize the importance of safety in the workplace.

## CONSIDERING SAFETY FOR YOUR ROBOT INSTALLATION

Safety is essential whenever robots are used. Keep in mind the following factors with regard to safety:

- The safety of people and equipment
- Use of safety enhancing devices
- Techniques for safe teaching and manual operation of the robot(s)
- Techniques for safe automatic operation of the robot(s)
- Regular scheduled inspection of the robot and workcell
- Proper maintenance of the robot

## Keeping People Safe

The safety of people is always of primary importance in any situation. When applying safety measures to your robotic system, consider the following:

- External devices
- Robot(s)
- Tooling
- Workpiece

## Using Safety Enhancing Devices

Always give appropriate attention to the work area that surrounds the robot. The safety of the work area can be enhanced by the installation of some or all of the following devices:

- Safety fences, barriers, or chains
- Light curtains
- Interlocks
- Pressure mats
- Floor markings
- Warning lights
- Mechanical stops
- EMERGENCY STOP buttons
- DEADMAN switches

## Setting Up a Safe Workcell

A safe workcell is essential to protect people and equipment. Observe the following guidelines to ensure that the workcell is set up safely. These suggestions are intended to supplement and not replace existing federal, state, and local laws, regulations, and guidelines that pertain to safety.

- Sponsor your personnel for training in approved FANUC America Corporation training course(s) related to your application. Never permit untrained personnel to operate the robots.
- Install a lockout device that uses an access code to prevent unauthorized persons from operating the robot.
- Use anti-tie-down logic to prevent the operator from bypassing safety measures.
- Arrange the workcell so the operator faces the workcell and can see what is going on inside the cell.
- Clearly identify the work envelope of each robot in the system with floor markings, signs, and special barriers. The work envelope is the area defined by the maximum motion range of the robot, including any tooling attached to the wrist flange that extend this range.



- Position all controllers outside the robot work envelope.
- Never rely on software or firmware based controllers as the primary safety element unless they comply with applicable current robot safety standards.
- Mount an adequate number of EMERGENCY STOP buttons or switches within easy reach of the operator and at critical points inside and around the outside of the workcell.
- Install flashing lights and/or audible warning devices that activate whenever the robot is operating, that is, whenever power is applied to the servo drive system. Audible warning devices shall exceed the ambient noise level at the end-use application.
- Wherever possible, install safety fences to protect against unauthorized entry by personnel into the work envelope.
- Install special guarding that prevents the operator from reaching into restricted areas of the work envelope.
- Use interlocks.
- Use presence or proximity sensing devices such as light curtains, mats, and capacitance and vision systems to enhance safety.
- Periodically check the safety joints or safety clutches that can be optionally installed between the robot wrist flange and tooling. If the tooling strikes an object, these devices dislodge, remove power from the system, and help to minimize damage to the tooling and robot.
- Make sure all external devices are properly filtered, grounded, shielded, and suppressed to prevent hazardous motion due to the effects of electro-magnetic interference (EMI), radio frequency interference (RFI), and electro-static discharge (ESD).
- Make provisions for power lockout/tagout at the controller.
- Eliminate *pinch points*. Pinch points are areas where personnel could get trapped between a moving robot and other equipment.
- Provide enough room inside the workcell to permit personnel to teach the robot and perform maintenance safely.
- Program the robot to load and unload material safely.
- If high voltage electrostatics are present, be sure to provide appropriate interlocks, warning, and beacons.
- If materials are being applied at dangerously high pressure, provide electrical interlocks for lockout of material flow and pressure.

### **Staying Safe While Teaching or Manually Operating the Robot**

Advise all personnel who must teach the robot or otherwise manually operate the robot to observe the following rules:

- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- Know whether or not you are using an intrinsically safe teach pendant if you are working in a hazardous environment.

- Before teaching, visually inspect the robot and work envelope to make sure that no potentially hazardous conditions exist. The work envelope is the area defined by the maximum motion range of the robot. These include tooling attached to the wrist flange that extends this range.
- The area near the robot must be clean and free of oil, water, or debris. Immediately report unsafe working conditions to the supervisor or safety department.
- FANUC America Corporation recommends that no one enter the work envelope of a robot that is on, except for robot teaching operations. However, if you must enter the work envelope, be sure all safeguards are in place, check the teach pendant DEADMAN switch for proper operation, and place the robot in teach mode. Take the teach pendant with you, turn it on, and be prepared to release the DEADMAN switch. Only the person with the teach pendant should be in the work envelope.

 **WARNING**

**Never bypass, strap, or otherwise deactivate a safety device, such as a limit switch, for any operational convenience. Deactivating a safety device is known to have resulted in serious injury and death.**

- Know the path that can be used to escape from a moving robot; make sure the escape path is never blocked.
- Isolate the robot from all remote control signals that can cause motion while data is being taught.
- Test any program being run for the first time in the following manner:

 **WARNING**

**Stay outside the robot work envelope whenever a program is being run. Failure to do so can result in injury.**

- Using a low motion speed, single step the program for at least one full cycle.
  - Using a low motion speed, test run the program continuously for at least one full cycle.
  - Using the programmed speed, test run the program continuously for at least one full cycle.
- Make sure all personnel are outside the work envelope before running production.

## **Staying Safe During Automatic Operation**

Advise all personnel who operate the robot during production to observe the following rules:

- Make sure all safety provisions are present and active.

- Know the entire workcell area. The workcell includes the robot and its work envelope, plus the area occupied by all external devices and other equipment with which the robot interacts.
- Understand the complete task the robot is programmed to perform before initiating automatic operation.
- Make sure all personnel are outside the work envelope before operating the robot.
- Never enter or allow others to enter the work envelope during automatic operation of the robot.
- Know the location and status of all switches, sensors, and control signals that could cause the robot to move.
- Know where the EMERGENCY STOP buttons are located on both the robot control and external control devices. Be prepared to press these buttons in an emergency.
- Never assume that a program is complete if the robot is not moving. The robot could be waiting for an input signal that will permit it to continue its activity.
- If the robot is running in a pattern, do not assume it will continue to run in the same pattern.
- Never try to stop the robot, or break its motion, with your body. The only way to stop robot motion immediately is to press an EMERGENCY STOP button located on the controller panel, teach pendant, or emergency stop stations around the workcell.

### **Staying Safe During Inspection**

When inspecting the robot, be sure to

- Turn off power at the controller.
- Lock out and tag out the power source at the controller according to the policies of your plant.
- Turn off the compressed air source and relieve the air pressure.
- If robot motion is not needed for inspecting the electrical circuits, press the EMERGENCY STOP button on the operator panel.
- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- If power is needed to check the robot motion or electrical circuits, be prepared to press the EMERGENCY STOP button, in an emergency.
- Be aware that when you remove a servomotor or brake, the associated robot arm will fall if it is not supported or resting on a hard stop. Support the arm on a solid support before you release the brake.

### **Staying Safe During Maintenance**

When performing maintenance on your robot system, observe the following rules:

- Never enter the work envelope while the robot or a program is in operation.
- Before entering the work envelope, visually inspect the workcell to make sure no potentially hazardous conditions exist.

- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- Consider all or any overlapping work envelopes of adjoining robots when standing in a work envelope.
- Test the teach pendant for proper operation before entering the work envelope.
- If it is necessary for you to enter the robot work envelope while power is turned on, you must be sure that you are in control of the robot. Be sure to take the teach pendant with you, press the DEADMAN switch, and turn the teach pendant on. Be prepared to release the DEADMAN switch to turn off servo power to the robot immediately.
- Whenever possible, perform maintenance with the power turned off. Before you open the controller front panel or enter the work envelope, turn off and lock out the 3-phase power source at the controller.
- Be aware that when you remove a servomotor or brake, the associated robot arm will fall if it is not supported or resting on a hard stop. Support the arm on a solid support before you release the brake.

 **WARNING**

**Lethal voltage is present in the controller WHENEVER IT IS CONNECTED to a power source. Be extremely careful to avoid electrical shock. HIGH VOLTAGE IS PRESENT at the input side whenever the controller is connected to a power source. Turning the disconnect or circuit breaker to the OFF position removes power from the output side of the device only.**

- Release or block all stored energy. Before working on the pneumatic system, shut off the system air supply and purge the air lines.
- Isolate the robot from all remote control signals. If maintenance must be done when the power is on, make sure the person inside the work envelope has sole control of the robot. The teach pendant must be held by this person.
- Make sure personnel cannot get trapped between the moving robot and other equipment. Know the path that can be used to escape from a moving robot. Make sure the escape route is never blocked.
- Use blocks, mechanical stops, and pins to prevent hazardous movement by the robot. Make sure that such devices do not create pinch points that could trap personnel.

 **WARNING**

**Do not try to remove any mechanical component from the robot before thoroughly reading and understanding the procedures in the appropriate manual. Doing so can result in serious personal injury and component destruction.**

- Be aware that when you remove a servomotor or brake, the associated robot arm will fall if it is not supported or resting on a hard stop. Support the arm on a solid support before you release the brake.
- When replacing or installing components, make sure dirt and debris do not enter the system.
- Use only specified parts for replacement. To avoid fires and damage to parts in the controller, never use nonspecified fuses.
- Before restarting a robot, make sure no one is inside the work envelope; be sure that the robot and all external devices are operating normally.

## KEEPING MACHINE TOOLS AND EXTERNAL DEVICES SAFE

Certain programming and mechanical measures are useful in keeping the machine tools and other external devices safe. Some of these measures are outlined below. Make sure you know all associated measures for safe use of such devices.

### Programming Safety Precautions

Implement the following programming safety measures to prevent damage to machine tools and other external devices.

- Back-check limit switches in the workcell to make sure they do not fail.
- Implement “failure routines” in programs that will provide appropriate robot actions if an external device or another robot in the workcell fails.
- Use *handshaking* protocol to synchronize robot and external device operations.
- Program the robot to check the condition of all external devices during an operating cycle.

### Mechanical Safety Precautions

Implement the following mechanical safety measures to prevent damage to machine tools and other external devices.

- Make sure the workcell is clean and free of oil, water, and debris.
- Use DCS (Dual Check Safety), software limits, limit switches, and mechanical hardstops to prevent undesired movement of the robot into the work area of machine tools and external devices.

## KEEPING THE ROBOT SAFE

Observe the following operating and programming guidelines to prevent damage to the robot.

### Operating Safety Precautions

The following measures are designed to prevent damage to the robot during operation.

- Use a low override speed to increase your control over the robot when jogging the robot.
- Visualize the movement the robot will make before you press the jog keys on the teach pendant.
- Make sure the work envelope is clean and free of oil, water, or debris.
- Use circuit breakers to guard against electrical overload.

### Programming Safety Precautions

The following safety measures are designed to prevent damage to the robot during programming:

- Establish *interference zones* to prevent collisions when two or more robots share a work area.
- Make sure that the program ends with the robot near or at the home position.
- Be aware of signals or other operations that could trigger operation of tooling resulting in personal injury or equipment damage.
- In dispensing applications, be aware of all safety guidelines with respect to the dispensing materials.

**NOTE:** Any deviation from the methods and safety practices described in this manual must conform to the approved standards of your company. If you have questions, see your supervisor.

## ADDITIONAL SAFETY CONSIDERATIONS FOR PAINT ROBOT INSTALLATIONS

Process technicians are sometimes required to enter the paint booth, for example, during daily or routine calibration or while teaching new paths to a robot. Maintenance personnel also must work inside the paint booth periodically.

Whenever personnel are working inside the paint booth, ventilation equipment must be used. Instruction on the proper use of ventilating equipment usually is provided by the paint shop supervisor.

Although paint booth hazards have been minimized, potential dangers still exist. Therefore, today's highly automated paint booth requires that process and maintenance personnel have full awareness of the system and its capabilities. They must understand the interaction that occurs between the vehicle moving along the conveyor and the robot(s), hood/deck and door opening devices, and high-voltage electrostatic tools.



### CAUTION

**Ensure that all ground cables remain connected. Never operate the paint robot with ground provisions disconnected. Otherwise, you could injure personnel or damage equipment.**

Paint robots are operated in three modes:

- Teach or manual mode
- Automatic mode, including automatic and exercise operation
- Diagnostic mode

During both teach and automatic modes, the robots in the paint booth will follow a predetermined pattern of movements. In teach mode, the process technician teaches (programs) paint paths using the teach pendant.

In automatic mode, robot operation is initiated at the System Operator Console (SOC) or Manual Control Panel (MCP), if available, and can be monitored from outside the paint booth. All personnel must remain outside of the booth or in a designated safe area within the booth whenever automatic mode is initiated at the SOC or MCP.

In automatic mode, the robots will execute the path movements they were taught during teach mode, but generally at production speeds.

When process and maintenance personnel run diagnostic routines that require them to remain in the paint booth, they must stay in a designated safe area.

## Paint System Safety Features

Process technicians and maintenance personnel must become totally familiar with the equipment and its capabilities. To minimize the risk of injury when working near robots and related equipment, personnel must comply strictly with the procedures in the manuals.

This section provides information about the safety features that are included in the paint system and also explains the way the robot interacts with other equipment in the system.

The paint system includes the following safety features:

- Most paint booths have red warning beacons that illuminate when the robots are armed and ready to paint. Your booth might have other kinds of indicators. Learn what these are.

- Some paint booths have a blue beacon that, when illuminated, indicates that the electrostatic devices are enabled. Your booth might have other kinds of indicators. Learn what these are.
- EMERGENCY STOP buttons are located on the robot controller and teach pendant. Become familiar with the locations of all E–STOP buttons.
- An intrinsically safe teach pendant is used when teaching in hazardous paint atmospheres.
- A DEADMAN switch is located on each teach pendant. When this switch is held in, and the teach pendant is on, power is applied to the robot servo system. If the engaged DEADMAN switch is released or pressed harder during robot operation, power is removed from the servo system, all axis brakes are applied, and the robot comes to an EMERGENCY STOP. Safety interlocks within the system might also E–STOP other robots.

 **WARNING**

**An EMERGENCY STOP will occur if the DEADMAN switch is released on a bypassed robot.**

- Overtravel by robot axes is prevented by software limits. All of the major and minor axes are governed by software limits. DCS (Dual Check Safety), limit switches and hardstops also limit travel by the major axes.
- EMERGENCY STOP limit switches and photoelectric eyes might be part of your system. Limit switches, located on the entrance/exit doors of each booth, will EMERGENCY STOP all equipment in the booth if a door is opened while the system is operating in automatic or manual mode. For some systems, signals to these switches are inactive when the switch on the SOC is in teach mode.
- When present, photoelectric eyes are sometimes used to monitor unauthorized intrusion through the entrance/exit silhouette openings.
- System status is monitored by computer. Severe conditions result in automatic system shutdown.

## **Staying Safe While Operating the Paint Robot**

When you work in or near the paint booth, observe the following rules, in addition to all rules for safe operation that apply to all robot systems.

 **WARNING**

**Observe all safety rules and guidelines to avoid injury.**



 **WARNING**

**Never bypass, strap, or otherwise deactivate a safety device, such as a limit switch, for any operational convenience. Deactivating a safety device is known to have resulted in serious injury and death.**

 **WARNING**

**Enclosures shall not be opened unless the area is known to be nonhazardous or all power has been removed from devices within the enclosure. Power shall not be restored after the enclosure has been opened until all combustible dusts have been removed from the interior of the enclosure and the enclosure purged. Refer to the Purge chapter for the required purge time.**

- Know the work area of the entire paint station (workcell).
- Know the work envelope of the robot and hood/deck and door opening devices.
- Be aware of overlapping work envelopes of adjacent robots.
- Know where all red, mushroom-shaped EMERGENCY STOP buttons are located.
- Know the location and status of all switches, sensors, and/or control signals that might cause the robot, conveyor, and opening devices to move.
- Make sure that the work area near the robot is clean and free of water, oil, and debris. Report unsafe conditions to your supervisor.
- Become familiar with the complete task the robot will perform BEFORE starting automatic mode.
- Make sure all personnel are outside the paint booth before you turn on power to the robot servo system.
- Never enter the work envelope or paint booth before you turn off power to the robot servo system.
- Never enter the work envelope during automatic operation unless a safe area has been designated.
- Never wear watches, rings, neckties, scarves, or loose clothing that could get caught in moving machinery.
- Remove all metallic objects, such as rings, watches, and belts, before entering a booth when the electrostatic devices are enabled.
- Stay out of areas where you might get trapped between a moving robot, conveyor, or opening device and another object.
- Be aware of signals and/or operations that could result in the triggering of guns or bells.
- Be aware of all safety precautions when dispensing of paint is required.
- Follow the procedures described in this manual.

## Special Precautions for Combustible Dusts (Powder Paint)

When the robot is used in a location where combustible dusts are found, such as the application of powder paint, the following special precautions are required to insure that there are no combustible dusts inside the robot.

- Purge maintenance air should be maintained at all times, even when the robot power is off. This will insure that dust can not enter the robot.
- A purge cycle will not remove accumulated dusts. Therefore, if the robot is exposed to dust when maintenance air is not present, it will be necessary to remove the covers and clean out any accumulated dust. Do not energize the robot until you have performed the following steps.
  1. Before covers are removed, the exterior of the robot should be cleaned to remove accumulated dust.
  2. When cleaning and removing accumulated dust, either on the outside or inside of the robot, be sure to use methods appropriate for the type of dust that exists. Usually lint free rags dampened with water are acceptable. Do not use a vacuum cleaner to remove dust as it can generate static electricity and cause an explosion unless special precautions are taken.
  3. Thoroughly clean the interior of the robot with a lint free rag to remove any accumulated dust.
  4. When the dust has been removed, the covers must be replaced immediately.
  5. Immediately after the covers are replaced, run a complete purge cycle. The robot can now be energized.

## Staying Safe While Operating Paint Application Equipment

When you work with paint application equipment, observe the following rules, in addition to all rules for safe operation that apply to all robot systems.



### WARNING

**When working with electrostatic paint equipment, follow all national and local codes as well as all safety guidelines within your organization. Also reference the following standards: NFPA 33 Standards for Spray Application Using Flammable or Combustible Materials, and NFPA 70 National Electrical Code.**

- **Grounding:** All electrically conductive objects in the spray area must be grounded. This includes the spray booth, robots, conveyors, workstations, part carriers, hooks, paint pressure pots, as well as solvent containers. Grounding is defined as the object or objects shall be electrically connected to ground with a resistance of not more than 1 megohms.
- **High Voltage:** High voltage should only be on during actual spray operations. Voltage should be off when the painting process is completed. Never leave high voltage on during a cap cleaning process.
- Avoid any accumulation of combustible vapors or coating matter.
- Follow all manufacturer recommended cleaning procedures.
- Make sure all interlocks are operational.

- No smoking.
- Post all warning signs regarding the electrostatic equipment and operation of electrostatic equipment according to NFPA 33 Standard for Spray Application Using Flammable or Combustible Material.
- Disable all air and paint pressure to bell.
- Verify that the lines are not under pressure.

### Staying Safe During Maintenance

When you perform maintenance on the painter system, observe the following rules, and all other maintenance safety rules that apply to all robot installations. Only qualified, trained service or maintenance personnel should perform repair work on a robot.

- Paint robots operate in a potentially explosive environment. Use caution when working with electric tools.
- When a maintenance technician is repairing or adjusting a robot, the work area is under the control of that technician. All personnel not participating in the maintenance must stay out of the area.
- For some maintenance procedures, station a second person at the control panel within reach of the EMERGENCY STOP button. This person must understand the robot and associated potential hazards.
- Be sure all covers and inspection plates are in good repair and in place.
- Always return the robot to the “home” position before you disarm it.
- Never use machine power to aid in removing any component from the robot.
- During robot operations, be aware of the robot’s movements. Excess vibration, unusual sounds, and so forth, can alert you to potential problems.
- Whenever possible, turn off the main electrical disconnect before you clean the robot.
- When using vinyl resin observe the following:
  - Wear eye protection and protective gloves during application and removal.
  - Adequate ventilation is required. Overexposure could cause drowsiness or skin and eye irritation.
  - If there is contact with the skin, wash with water.
  - Follow the Original Equipment Manufacturer’s Material Safety Data Sheets.
- When using paint remover observe the following:
  - Eye protection, protective rubber gloves, boots, and apron are required during booth cleaning.
  - Adequate ventilation is required. Overexposure could cause drowsiness.
  - If there is contact with the skin or eyes, rinse with water for at least 15 minutes. Then seek medical attention as soon as possible.
  - Follow the Original Equipment Manufacturer’s Material Safety Data Sheets.



# SAFETY PRECAUTIONS

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This chapter describes the precautions which must be followed to ensure the safe use of the robot. Before using the robot, be sure to read this chapter thoroughly.

For detailed functions of the robot operation, read the relevant operator's manual to understand fully its specification.

For the safety of the operator and the system, follow all safety precautions when operating a robot and its peripheral equipment installed in a work cell.

In addition, refer to the "FANUC Robot SAFETY HANDBOOK (B-80687EN)".

## 1 DEFINITION OF USER

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The user can be defined as follows.

**Operator:**

- Turns ON/OFF power to the robot
- Starts the robot program from the operator's panel

**Programmer:**

- Operates the robot
- Teaches the robot inside the safety fence

**Maintenance engineer:**

- Operates the robot
- Teaches the robot inside the safety fence
- Performs maintenance (repair, adjustment, replacement)

- Operator is not allowed to work in the safety fence.
- Programmers and maintenance engineers are allowed to work in the safety fence. The work inside the safety fence includes lifting, setting, teaching, adjustment, maintenance, etc.
- To work inside the safety fence, the person must receive a professional training for the robot.

During the operation, programming, and maintenance of your robotic system, the programmer, operator, and maintenance engineer should take additional care of their safety by wearing the following safety items.

- Adequate clothes for the operation
- Safety shoes
- A helmet

## 2 DEFINITION OF SAFETY NOTATIONS

To ensure the safety of users and prevent damage to the machine, this manual indicates each precaution on safety with "**WARNING**" or "**CAUTION**" according to its severity. Supplementary information is indicated by "**NOTE**". Read the contents of each "**WARNING**", "**CAUTION**" and "**NOTE**" before using the robot.

| Symbol                                                                                           | Definitions                                                                                                                                                               |
|--------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <b>WARNING</b> | Used if hazard resulting in the death or serious injury of the user will be expected to occur if he or she fails to follow the approved procedure.                        |
|  <b>CAUTION</b> | Used if a hazard resulting in the minor or moderate injury of the user, or equipment damage may be expected to occur if he or she fails to follow the approved procedure. |
| <b>NOTE</b>                                                                                      | Used if a supplementary explanation not related to any of WARNING and CAUTION is to be indicated.                                                                         |

- Check this manual thoroughly, and keep it handy for the future reference.

## 3 SAFETY OF THE USER

User safety is the primary safety consideration. Because it is very dangerous to enter the operating space of the robot during automatic operation, adequate safety precautions must be observed.

The following lists the general safety precautions. Careful consideration must be made to ensure user safety.

- (1) Have the robot system users attend the training courses held by FANUC.

FANUC provides various training courses. Contact our sales office for details.

- (2) Even when the robot is stationary, it is possible that the robot is still in a ready to move state, and is waiting for a signal. In this state, the robot is regarded as still in motion. To ensure user safety, provide the system with an alarm to indicate visually or aurally that the robot is in motion.
- (3) Install a safety fence with a gate so that no user can enter the work area without passing through the gate. Install an interlocking device, a safety plug, and so forth in the safety gate so that the robot is stopped as the safety gate is opened.

The controller is designed to receive this interlocking signal of the door switch. When the gate is opened and this signal received, the controller stops the robot (Please refer to "STOP TYPE OF ROBOT" in "SAFETY PRECAUTIONS" for detail of stop type). For connection, see Fig. 3 (b).

- (4) Provide the peripheral equipment with appropriate earth (Class A, Class B, Class C, and Class D).
- (5) Try to install the peripheral equipment outside the robot operating space.
- (6) Draw an outline on the floor, clearly indicating the range of the robot operating space, including the tools such as a hand.
- (7) Install a mat switch or photoelectric switch on the floor with an interlock to a visual or aural alarm that stops the robot when a user enters the work area.
- (8) If necessary, install a safety lock so that no one except the user in charge can turn on the power of the robot.

The circuit breaker installed in the controller is designed to disable anyone from turning it on when it is locked with a padlock.

- (9) When adjusting each peripheral equipment independently, be sure to turn off the power of the robot.
- (10) Operators should be ungloved while manipulating the operator panel or teach pendant. Operation with gloved fingers could cause an operation error.
- (11) Programs, system variables, and other information can be saved on memory card or USB memories. Be sure to save the data periodically in case the data is lost in an accident. (refer to Controller OPERATOR'S MANUAL.)
- (12) The robot should be transported and installed by accurately following the procedures recommended by FANUC. Wrong transportation or installation may cause the robot to fall, resulting in severe injury to workers.
- (13) In the first operation of the robot after installation, the operation should be restricted to low speeds. Then, the speed should be gradually increased to check the operation of the robot.
- (14) Before the robot is started, it should be checked that no one is inside the safety fence. At the same time, a check must be made to ensure that there is no risk of hazardous situations. If detected, such a situation should be eliminated before the operation.
- (15) When the robot is used, the following precautions should be taken. Otherwise, the robot and peripheral equipment can be adversely affected, or workers can be severely injured.
  - Avoid using the robot in a flammable environment.
  - Avoid using the robot in an explosive environment.
  - Avoid using the robot in an environment full of radiation.
  - Avoid using the robot under water or at high humidity.
  - Avoid using the robot to carry a person or animal.
  - Avoid using the robot as a stepladder. (Never climb up on or hang from the robot.)
  - Outdoor
- (16) When connecting the peripheral equipment related to stop (safety fence etc.) and each signal (external emergency, fence etc.) of robot, be sure to confirm the stop movement and do not take the wrong connection.
- (17) When preparing footstep, please consider security for installation and maintenance work in high place according to Fig. 3 (c). Please consider footstep and safety belt mounting position.

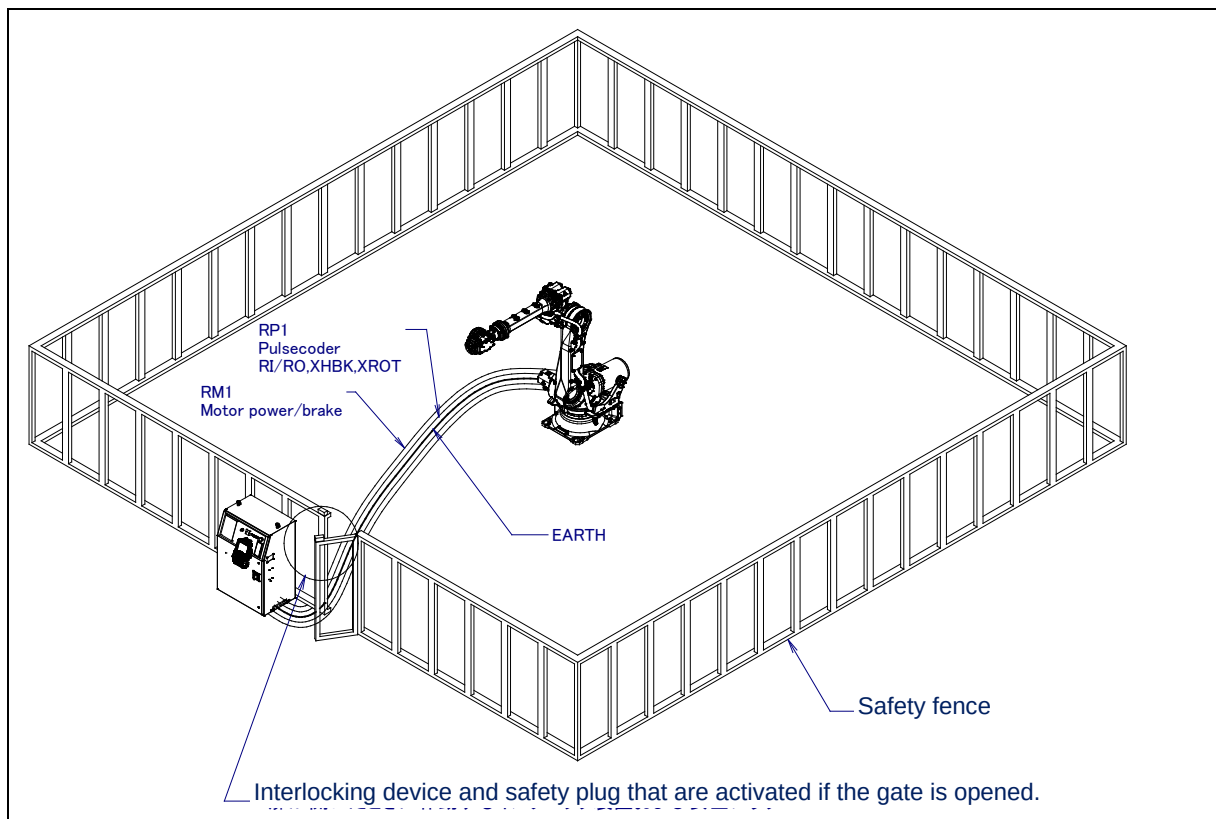


Fig. 3 (a) Safety fence and safety gate

**⚠ WARNING**

When you close a fence, please confirm that there is not a person from all directions of the robot.

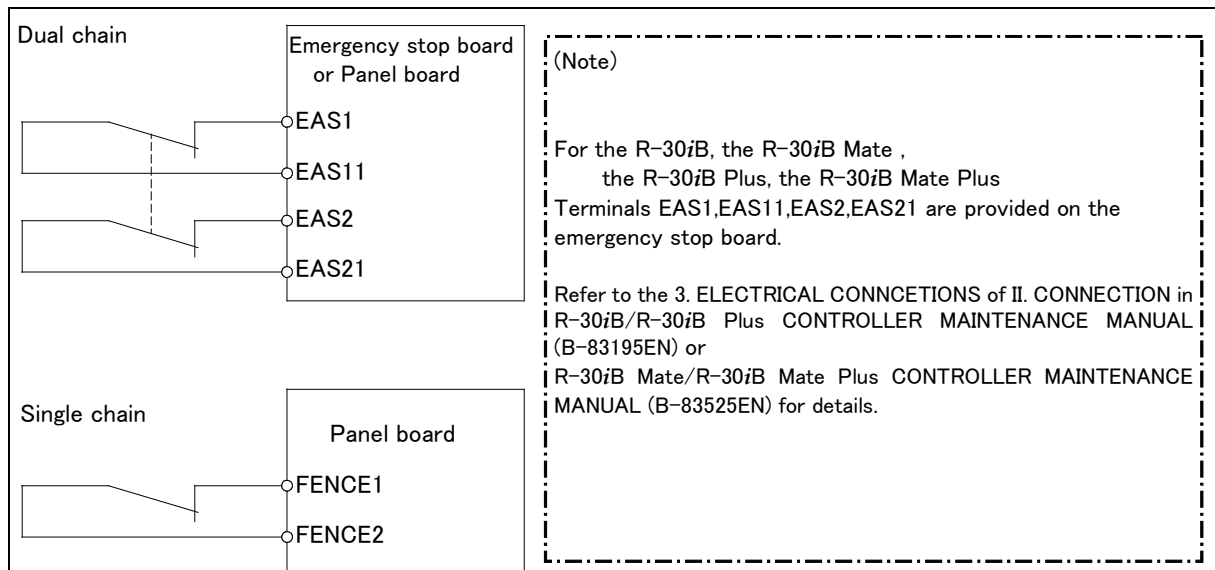


Fig. 3 (b) Connection diagram for the signal of safety fence

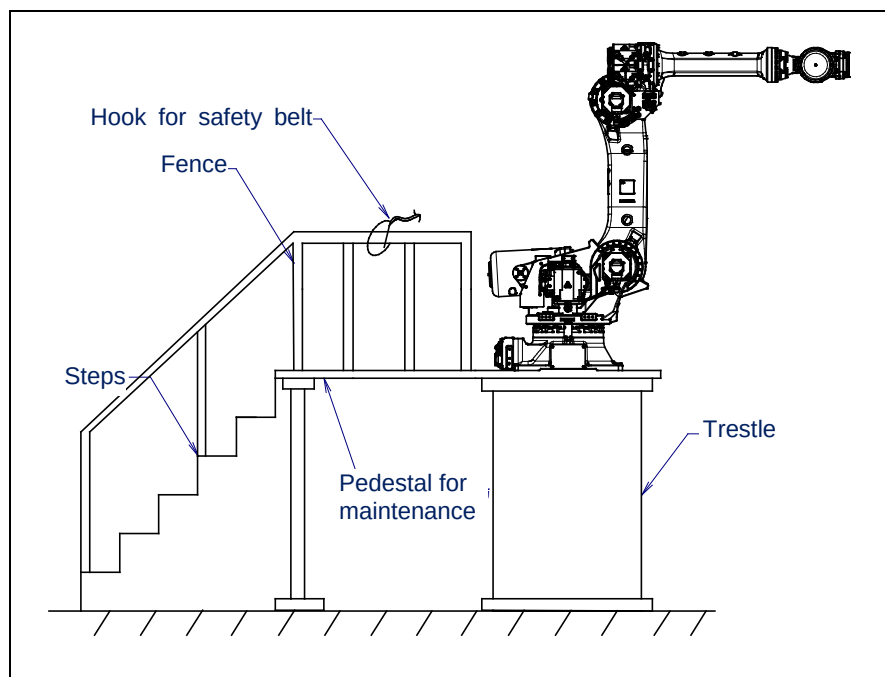


Fig. 3 (c) Pedestal for maintenance

## 3.1 SAFETY OF THE OPERATOR

An operator refers to a person who turns on and off the robot system and starts a robot program from, for example, the operator panel during daily operation.

Operators cannot work inside of the safety fence.

- (1) If the robot does not need to be operated, turn off the robot controller power or press the EMERGENCY STOP button during working.
- (2) Operate the robot system outside the operating space of the robot.



- (3) Install a safety fence or safety door to avoid the accidental entry of a person other than an operator in charge or keep operator out from the hazardous place.
- (4) Install one or more necessary quantity of EMERGENCY STOP button(s) within the operator's reach in appropriate location(s) based on the system layout.

The robot controller is designed to be connected to an external EMERGENCY STOP button. With this connection, the controller stops the robot operation (Please refer to "STOP TYPE OF ROBOT" in "SAFETY PRECAUTIONS" for detail of stop type) when the external EMERGENCY STOP button is pressed. See the diagram below for connection.

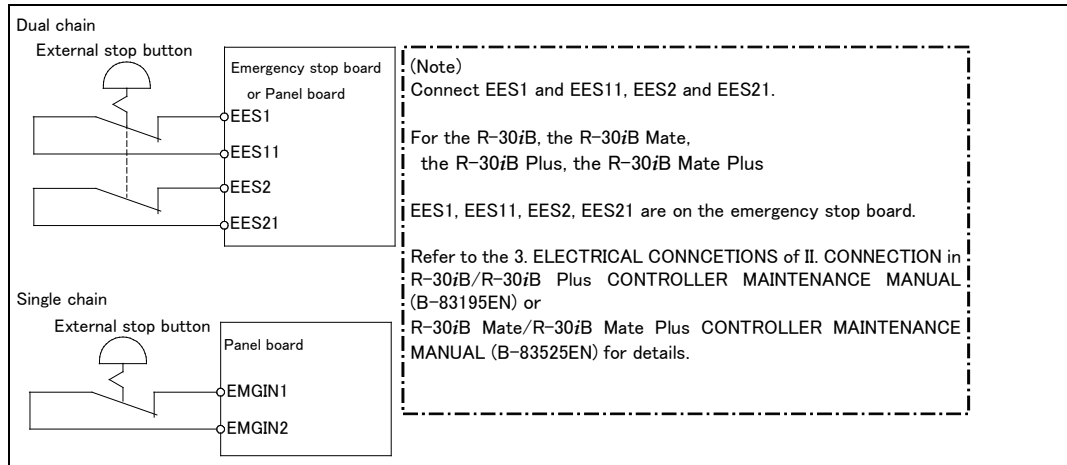


Fig. 3.1 Connection diagram for external emergency stop button

## 3.2 SAFETY OF THE PROGRAMMER

While teaching the robot, the operator may need to enter the robot operation area. The programmer must ensure the safety especially.

- (1) Unless it is specifically necessary to enter the robot operating space, carry out all tasks outside the operating space.
- (2) Before teaching the robot, check that the robot and its peripheral equipment are all in the normal operating condition.
- (3) If it is inevitable to enter the robot operating space to teach the robot, check the locations, settings, and other conditions of the safety devices (such as the EMERGENCY STOP button, the DEADMAN switch on the teach pendant) before entering the area.
- (4) The programmer must be extremely careful not to let anyone else enter the robot operating space.
- (5) Programming should be done outside the area of the safety fence as far as possible. If programming needs to be done inside the safety fence, the programmer should take the following precautions:
  - Before entering the area of the safety fence, ensure that there is no risk of dangerous situations in the area.
  - Be prepared to press the emergency stop button whenever necessary.
  - Robot motions should be made at low speeds.
  - Before starting programming, check the whole robot system status to ensure that no remote instruction to the peripheral equipment or motion would be dangerous to the user.

Our operator panel is provided with an emergency stop button and a key switch (mode switch) for selecting the automatic operation (AUTO) and the teach modes (T1 and T2). Before entering the inside of the safety fence for the purpose of teaching, set the switch to a teach mode, remove the key from the mode switch to prevent other people from changing the operation mode carelessly, then open the safety gate. If the safety gate is opened with the automatic operation set, the robot stops (Please refer to "STOP TYPE OF ROBOT" in "SAFETY PRECAUTIONS" for detail of stop type). After the switch is set to a teach mode, the safety gate is disabled. The programmer should understand that the safety gate is disabled and is responsible for keeping other people from entering the inside of the safety fence.

Our teach pendant is provided with a DEADMAN switch as well as an emergency stop button. These button and switch function as follows:

- (1) Emergency stop button: Causes the stop of the robot (Please refer to "STOP TYPE OF ROBOT" in "SAFETY PRECAUTIONS" for detail of stop type) when pressed.
  - (2) DEADMAN switch: Functions differently depending on the teach pendant enable/disable switch setting status.
    - (a) Enable: Servo power is turned off when the operator releases the DEADMAN switch or when the operator presses the switch strongly.
    - (b) Disable: The DEADMAN switch is disabled.
- (Note) The DEADMAN switch is provided to stop the robot when the operator releases the teach pendant or presses the pendant strongly in case of emergency. The R-30iB Plus/R-30iB Mate Plus employs a 3-position DEADMAN switch, which allows the robot to operate when the 3-position DEADMAN switch is pressed to its intermediate point. When the operator releases the DEADMAN switch or presses the switch strongly, the robot stops immediately.

The operator's intention of starting teaching is determined by the controller through the dual operation of setting the teach pendant enable/disable switch to the enable position and pressing the DEADMAN switch. The operator should make sure that the robot could operate in such conditions and be responsible in carrying out tasks safely.

Based on the risk assessment by FANUC, number of operation of DEADMAN SW should not exceed about 10000 times per year.

The teach pendant, operator panel, and peripheral equipment interface send each robot start signal. However the validity of each signal changes as follows depending on the mode switch and the DEADMAN switch of the operator panel, the teach pendant enable switch and the remote condition on the software.

| Mode        | Teach pendant enable switch | Software remote condition | Teach pendant    | Operator panel   | Peripheral equipment |
|-------------|-----------------------------|---------------------------|------------------|------------------|----------------------|
| AUTO mode   | On                          | Local                     | Not allowed      | Not allowed      | Not allowed          |
|             |                             | Remote                    | Not allowed      | Not allowed      | Not allowed          |
|             | Off                         | Local                     | Not allowed      | Allowed to start | Not allowed          |
|             |                             | Remote                    | Not allowed      | Not allowed      | Allowed to start     |
| T1, T2 mode | On                          | Local                     | Allowed to start | Not allowed      | Not allowed          |
|             |                             | Remote                    | Allowed to start | Not allowed      | Not allowed          |
|             | Off                         | Local                     | Not allowed      | Not allowed      | Not allowed          |
|             |                             | Remote                    | Not allowed      | Not allowed      | Not allowed          |

**T1,T2 mode: DEADMAN switch is effective.**

- (6) To start the system using the operator box or operator panel, make certain that nobody is the robot operating space area and that there are no abnormalities in the robot operating space.
- (7) When a program is completed, be sure to carry out a test operation according to the following procedure.
  - (a) Run the program for at least one operation cycle in the single step mode at low speed.
  - (b) Run the program for at least one operation cycle in continuous operation at low speed.
  - (c) Run the program for one operation cycle in continuous operation at the intermediate speed and check that no abnormalities occur due to a delay in timing.

- (d) Run the program for one operation cycle in continuous operation at the normal operating speed and check that the system operates automatically without trouble.
- (e) After checking the completeness of the program through the test operation above, execute it in the automatic operation.
- (8) While operating the system in the automatic operation, the programmer should leave the safety fence.

### 3.3 SAFETY OF THE MAINTENANCE ENGINEER

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For the safety of maintenance engineer personnel, pay utmost attention to the following.

- (1) During operation, never enter the robot operating space.
- (2) A hazardous situation may arise when the robot or the system, are kept with their power-on during maintenance operations. Therefore, for any maintenance operation, the robot and the system should be put into the power-off state. If necessary, a lock should be in place in order to prevent any other person from turning on the robot and/or the system. In case maintenance needs to be executed in the power-on state, the emergency stop button must be pressed as far as possible.
- (3) If it becomes necessary to enter the robot operating space while the power is on, press the emergency stop button on the operator box or operator panel, or the teach pendant before entering the range. The maintenance worker must indicate that maintenance work is in progress and be careful not to allow other people to operate the robot carelessly.
- (4) When entering the area enclosed by the safety fence, the worker must check the whole robot system in order to make sure no dangerous situations exist. In case the worker needs to enter the safety area whilst a dangerous situation exists, extreme care must be taken, and whole robot system status must be carefully monitored.
- (5) Before the maintenance of the pneumatic system is started, the supply pressure should be shut off and the pressure in the piping should be reduced to zero.
- (6) Before the start of maintenance work, check that the robot and its peripheral equipment are all in the normal operating condition.
- (7) Do not operate the robot in the automatic operation while anybody is in the robot operating space.
- (8) When you maintain the robot alongside a wall or instrument, or when multiple users are working nearby, make certain that their escape path is not obstructed.
- (9) When a tool is mounted on the robot, or when any movable device other than the robot is installed, such as belt conveyor, pay careful attention to its motion.
- (10) If necessary, have a user who is familiar with the robot system stand beside the operator panel and observe the work being performed. If any danger arises, the user should be ready to press the EMERGENCY STOP button at any time.
- (11) When replacing a part, please contact your local FANUC representative. If a wrong procedure is followed, an accident may occur, causing damage to the robot and injury to the user.
- (12) When replacing or reinstalling components, take care to prevent foreign material from entering the system.
- (13) When handling each unit or printed circuit board in the controller during inspection, turn off the circuit breaker to protect against electric shock.  
If there are two cabinets, turn off the both circuit breaker.
- (14) A part should be replaced with a part recommended by FANUC. If other parts are used, malfunction or damage would occur. Especially, a fuse that is not recommended by FANUC should not be used. Such a fuse may cause a fire.
- (15) When restarting the robot system after completing maintenance work, make sure in advance that there is no person in the operating space and that the robot and the peripheral equipment are not abnormal.
- (16) When a motor or brake is removed, the robot arm should be supported with a crane or other equipment beforehand so that the arm would not fall during the removal.
- (17) Whenever grease is spilled on the floor, it should be removed as quickly as possible to prevent dangerous falls.

- (18) The following parts are heated. If a maintenance user needs to touch such a part in the heated state, the user should wear heat-resistant gloves or use other protective tools.
  - Servo motor
  - Inside the controller
  - Reducer
  - Gearbox
  - Wrist unit
- (19) Maintenance should be done under suitable light. Care must be taken that the light would not cause any danger.
- (20) When a motor, reducer, or other heavy load is handled, a crane or other equipment should be used to protect maintenance workers from excessive load. Otherwise, the maintenance workers would be severely injured.
- (21) The robot should not be stepped on or climbed up during maintenance. If it is attempted, the robot would be adversely affected. In addition, a misstep can cause injury to the worker.
- (22) When performing maintenance work in high place, secure a footstep and wear safety belt.
- (23) After the maintenance is completed, spilled oil or water and metal chips should be removed from the floor around the robot and within the safety fence.
- (24) When a part is replaced, all bolts and other related components should put back into their original places. A careful check must be given to ensure that no components are missing or left not mounted.
- (25) In case robot motion is required during maintenance, the following precautions should be taken :
  - Foresee an escape route. And during the maintenance motion itself, monitor continuously the whole robot system so that your escape route will not become blocked by the robot, or by peripheral equipment.
  - Always pay attention to potentially dangerous situations, and be prepared to press the emergency stop button whenever necessary.
- (26) The robot should be periodically inspected. (Refer to the robot mechanical manual and controller maintenance manual.) A failure to do the periodical inspection can adversely affect the performance or service life of the robot and may cause an accident
- (27) After a part is replaced, a test execution should be given for the robot according to a predetermined method. (See TESTING section of “Controller operator’s manual”.) During the test execution, the maintenance worker should work outside the safety fence.

## **4 SAFETY OF THE TOOLS AND PERIPHERAL EQUIPMENT**

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### **4.1 PRECAUTIONS IN PROGRAMMING**

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- (1) Use a limit switch or other sensor to detect a dangerous condition and, if necessary, design the program to stop the robot when the sensor signal is received.
- (2) Design the program to stop the robot when an abnormality occurs in any other robots or peripheral equipment, even though the robot itself is normal.
- (3) For a system in which the robot and its peripheral equipment are in synchronous motion, particular care must be taken in programming so that they do not interfere with each other.
- (4) Provide a suitable interface between the robot and its peripheral equipment so that the robot can detect the states of all devices in the system and can be stopped according to the states.

### **4.2 PRECAUTIONS FOR MECHANISM**

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- (1) Keep the component cells of the robot system clean, operate the robot where insulated from the influence of oil, water, and dust.

- (2) Don't use unconfirmed liquid for cutting fluid and cleaning fluid.
- (3) Adopt limit switches or mechanical stoppers to limit the robot motion, and avoid the robot from collisions against peripheral equipment or tools.
- (4) Observe the following precautions about the mechanical unit cables. Failure to follow precautions may cause problems.
  - Use mechanical unit cable that have required user interface.
  - Do not add user cable or hose to inside of the mechanical unit.
  - Please do not obstruct the movement of the mechanical unit when cables are added to outside of mechanical unit.
  - In the case of the model that a cable is exposed, please do not perform remodeling (Adding a protective cover and fix an outside cable more) obstructing the behavior of the outcrop of the cable.
  - When installing user peripheral equipment on the robot mechanical unit, please pay attention that the device does not interfere with the robot itself.
- (5) The frequent power-off stop for the robot during operation causes the trouble of the robot. Please avoid the system construction that power-off stop would be operated routinely. (Refer to bad case example.) Please perform power-off stop after reducing the speed of the robot and stopping it by hold stop or cycle stop when it is not urgent. (Please refer to "STOP TYPE OF ROBOT" in "SAFETY PRECAUTIONS" for detail of stop type.)  
(Bad case example)
  - Whenever poor product is generated, a line stops by emergency stop and power-off of the robot is incurred.
  - When alteration is necessary, safety switch is operated by opening safety fence and power-off stop is incurred for the robot during operation.
  - An operator pushes the emergency stop button frequently, and a line stops.
  - An area sensor or a mat switch connected to safety signal operates routinely and power-off stop is incurred for the robot.
  - Power-off stop is regularly incurred due to an inappropriate setting for Dual Check Safety (DCS).
- (6) Power-off stop of Robot is executed when collision detection alarm (SRVO-050) etc. occurs. Please try to avoid unnecessary power-off stops. It may cause the trouble of the robot, too. So remove the causes of the alarm.

## 5 SAFETY OF THE ROBOT MECHANICAL UNIT

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### 5.1 PRECAUTIONS IN OPERATION

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- (1) When operating the robot in the jog mode, set it at an appropriate speed so that the operator can manage the robot in any eventuality.
- (2) Before pressing the jog key, be sure you know in advance what motion the robot will perform in the jog mode.

### 5.2 PRECAUTIONS IN PROGRAMMING

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- (1) When the operating spaces of robots overlap, make certain that the motions of the robots do not interfere with each other.
- (2) Be sure to specify the predetermined work origin in a motion program for the robot and program the motion so that it starts from the origin and terminates at the origin. Make it possible for the operator to easily distinguish at a glance that the robot motion has terminated.

## 5.3 PRECAUTIONS FOR MECHANISMS

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- (1) Keep the robot operation area clean, and operate the robot in an environment free of grease, water, and dust.

## 5.4 PROCEDURE TO MOVE ARM WITHOUT DRIVE POWER IN EMERGENCY OR ABNORMAL SITUATIONS

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For emergency or abnormal situations (e.g. persons trapped in or pinched by the robot), brake release unit can be used to move the robot axes without drive power.

Please refer to controller maintenance manual and mechanical unit operator's manual for using method of brake release unit and method of supporting robot.

# 6 SAFETY OF THE END EFFECTOR

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## 6.1 PRECAUTIONS IN PROGRAMMING

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- (1) To control the pneumatic, hydraulic and electric actuators, carefully consider the necessary time delay after issuing each control command up to actual motion and ensure safe control.
- (2) Provide the end effector with a limit switch, and control the robot system by monitoring the state of the end effector.

# 7 STOP TYPE OF ROBOT (R-30iB Plus, R-30iB Mate Plus)

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There are following three types of Stop Category.

### Stop Category 0 following IEC 60204-1 (Power-off Stop)

Servo power is turned off, and the robot stops immediately. Servo power is turned off when the robot is moving, and the motion path of the deceleration is uncontrolled.

“Stop Category 0” performs following processing.

- An alarm is generated, and then the servo power turns off. Instantly the robot stops.
- Execution of the program is paused.

Frequent Category 0 Stop of the robot during operation can cause mechanical problems of the robot. Avoid system designs that require routine or frequent Category 0 Stop conditions.

### Stop Category 1 following IEC 60204-1 (Controlled Stop, Smooth Stop)

The robot is decelerated until it stops, and servo power is turned off.

“Stop Category 1” performs following processing.

- The alarm "SRVO-199 Controlled stop" or "SRVO-289 Smooth Stop" occurs along with a decelerated stop. The program execution is paused.
- An alarm is generated, and then the servo power turns off.

In Smooth stop, the robot decelerates until it stops with the deceleration time shorter than Controlled stop. The stop type of Stop Category 1 is different according to the robot model or option configuration. Please refer to the operator's manual of a particular robot model.

## Stop Category 2 following IEC 60204-1 (Hold)

The robot is decelerated until it stops, and servo power remains on.

“Stop Category 2” performs following processing.

- The robot operation is decelerated until it stops. Execution of the program is paused.

### WARNING

- 1 The stopping distance and time of Stop Category 1 are longer than those of Stop Category 0. A risk assessment for the whole robot system which takes into consideration the increased stopping distance and stopping time is necessary when Stop Category 1 is used. Please refer to the operator's manual of a particular robot model for the data of stopping distance and time.
- 2 In multi arm system, the longest stopping distance and time of Stop Category 1 among each robot are adopted as those for the system. A risk assessment for the whole robot system which takes into consideration a possibility that the stopping distance and time increase, is necessary on the multi arm system.
- 3 In the system which has extended axis, the longer stopping distance and time of Stop Category 1 among robot and extended axis are adopted as those for the system. A risk assessment for the whole robot system which takes into consideration a possibility that the stopping distance and time increase, is necessary on the system which has extended axis. Please refer to the extended axis setup procedure of the controller operator's manual for considering the stopping distance and time of the extended axis.
- 4 When Stop Category 1 occurs during deceleration by Stop Category 2, the stop type of robot is changed to Stop Category 0.
- 5 In case of Stop Category 1, motor power shutdown is delayed for a maximum of 2 seconds. In this case, a risk assessment for the whole robot system is necessary, including the 2 seconds delay.

When the emergency stop button is pressed or the FENCE is open, the stop type of robot is Stop Category 0 or Stop Category 1. The configuration of stop type for each situation is called *stop pattern*. The stop pattern is different according to the option configuration.

There are the following 3 Stop patterns.

| Stop pattern | Mode | Emergency stop button | External Emergency stop | FENCE open | SVOFF input | Deadman switch (*) |
|--------------|------|-----------------------|-------------------------|------------|-------------|--------------------|
| A            | AUTO | Category 0            | Category 0              | Category 1 | Category 1  | -                  |
|              | T1   | Category 0            | Category 0              | -          | Category 1  | Category 0         |
|              | T2   | Category 0            | Category 0              | -          | Category 1  | Category 0         |
| C            | AUTO | Category 1            | Category 1              | Category 1 | Category 1  | -                  |
|              | T1   | Category 0            | Category 0              | -          | Category 1  | Category 0         |
|              | T2   | Category 0            | Category 0              | -          | Category 1  | Category 0         |
| D            | AUTO | Category 1            | Category 1              | Category 1 | Category 1  | -                  |
|              | T1   | Category 1            | Category 1              | -          | Category 1  | Category 1         |
|              | T2   | Category 1            | Category 1              | -          | Category 1  | Category 1         |

Category 0: Stop Category 0

Category 1: Stop Category 1

-: Disable

(\*) The stop pattern of NTED input is same as Deadman switch.

The following table indicates the Stop pattern according to the controller type or option configuration. The case R651 is specified.

| Option                                    | R-30iB Plus/ R-30iB Mate Plus |
|-------------------------------------------|-------------------------------|
| Standard                                  | C(**)                         |
| Old Stop Function (A05B-2670-J680)        | A(**)                         |
| All Smooth Stop Function (A05B-2670-J651) | D(**)                         |

The case R650 is specified.

| Option                                     | R-30iB Plus/ R-30iB Mate Plus |
|--------------------------------------------|-------------------------------|
| Standard                                   | A(**)                         |
| Stop Category 1 by E-Stop (A05B-2670-J521) | C(**)                         |
| All Smooth Stop Function (A05B-2670-J651)  | D(**)                         |

(\*\*)R-30iB Mate Plus does not have SVOFF input.

The stop pattern of the controller is displayed in "Stop pattern" line in software version screen. Please refer to "Software version" in operator's manual of controller for the detail of software version screen.

### "Old Stop Function" option

When "**Old Stop Function**" (A05B-2670-J680) option is specified, the stop type of the following alarms becomes Stop Category 0 in AUTO mode.

| Alarm                               | Condition                                                       |
|-------------------------------------|-----------------------------------------------------------------|
| SRVO-001 Operator panel E-stop      | Operator panel emergency stop is pressed.                       |
| SRVO-002 Teach pendant E-stop       | Teach pendant emergency stop is pressed.                        |
| SRVO-007 External emergency stops   | External emergency stop input (EES1-EES11, EES2-EES21) is open. |
| SRVO-408 DCS SSO Ext Emergency Stop | In DCS Safe I/O connect function, SSO[3] is OFF.                |
| SRVO-409 DCS SSO Servo Disconnect   | In DCS Safe I/O connect function, SSO[4] is OFF.                |

**Stop Category 0** is different from **Stop Category 1** as follows:

- In Stop Category 0, servo power is turned off, and the robot stops immediately. Servo power is turned off when the robot is moving, and the motion path of the deceleration is uncontrolled.
- The stopping distance and time of Stop Category 0 is shorter than those of Stop Category 1, depending on the robot model and axis.

When this option is loaded, this function cannot be disabled.

The stop type of DCS Position and Speed Check functions is not affected by the loading of this option.

### "All Smooth Stop Function" option

When "**All Smooth Stop Function**" (A05B-2670-J651) option is specified, the stop type of the following alarms becomes Stop Category 1 in all operation modes (AUTO, T1 and T2 mode).

| Alarm                               | Condition                                                            |
|-------------------------------------|----------------------------------------------------------------------|
| SRVO-001 Operator panel E-stop      | Operator panel emergency stop is pressed.                            |
| SRVO-002 Teach pendant E-stop       | Teach pendant emergency stop is pressed.                             |
| SRVO-003 Deadman switch released    | Both deadman switches on Teach pendant are released.                 |
| SRVO-007 External emergency stops   | External emergency stop input (EES1-EES11, EES2-EES21) is open.      |
| SRVO-037 IMSTP input (Group: %d)    | IMSTP input (*IMSTP signal for a peripheral device interface) is ON. |
| SRVO-232 NTED input                 | NTED input (NTED1-NTED11, NTED2-NTED21) is open.                     |
| SRVO-408 DCS SSO Ext Emergency Stop | In DCS Safe I/O connect function, SSO[3] is OFF.                     |
| SRVO-409 DCS SSO Servo Disconnect   | In DCS Safe I/O connect function, SSO[4] is OFF.                     |
| SRVO-410 DCS SSO Ext Emergency Stop | In DCS Safe I/O connect function, SSO[5] is OFF.                     |
| SRVO-419 DCS PROFIsafe comm. error  | PROFINET Safety communication error occurs.                          |



**Stop Category 1** is different from **Stop Category 0** as follows:

- In Stop Category 1, the robot is stopped along the program path. This function is effective for a system where the robot can interfere with other devices if it deviates from the program path.
- In Stop Category 1, physical impact is less than Stop Category 0. This function is effective for systems where the physical impact to the mechanical unit or EOAT (End of Arm Tool) should be minimized.
- The stopping distance and time of Stop Category 1 is longer than those of Stop Category 0, depending on the robot model and axis.

When this option is loaded, this function cannot be disabled.

The stop type of DCS Position and Speed Check functions is not affected by the loading of this option.



#### **WARNING**

The stopping distance and time of Stop Category 1 are longer than those of Stop Category 0. A risk assessment for the whole robot system which takes into consideration the increased stopping distance and stopping time, is necessary when this option is loaded.

### **"Stop Category 1 by E-Stop" option**

When "**Stop Category 1 by E-Stop**" (A05B-2670-J521) option is specified, the stop type of the following alarms become Category 1 Stop but only in AUTO mode. In T1 or T2 mode, the stop type is Category 0 Stop which is the normal operation of the system.

| Alarm                               | Condition                                                       |
|-------------------------------------|-----------------------------------------------------------------|
| SRVO-001 Operator panel E-stop      | Operator panel emergency stop is pressed.                       |
| SRVO-002 Teach pendant E-stop       | Teach pendant emergency stop is pressed.                        |
| SRVO-007 External emergency stops   | External emergency stop input (EES1-EES11, EES2-EES21) is open. |
| SRVO-408 DCS SSO Ext Emergency Stop | In DCS Safe I/O connect function, SSO[3] is OFF.                |
| SRVO-409 DCS SSO Servo Disconnect   | In DCS Safe I/O connect function, SSO[4] is OFF.                |

**Stop Category 1** is different from **Stop Category 0** as follows:

- In Stop Category 1, the robot is stopped along the program path. This function is effective for a system where the robot can interfere with other devices if it deviates from the program path.
- In Stop Category 1, physical impact is less than Stop Category 0. This function is effective for systems where the physical impact to the mechanical unit or EOAT (End of Arm Tool) should be minimized.
- The stopping distance and time of Stop Category 1 is longer than those of Stop Category 0, depending on the robot model and axis.

When this option is loaded, this function cannot be disabled.

The stop type of DCS Position and Speed Check functions is not affected by the loading of this option.



#### **WARNING**

The stopping distance and time of Stop Category 1 are longer than those of Stop Category 0. A risk assessment for the whole robot system which takes into consideration the increased stopping distance and stopping time, is necessary when this option is loaded.



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# 1 ABOUT THIS MANUAL

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This manual is made from R-30iA/R-30iA Mate/R-30iB CONTROLLER CC-Link Interface (Slave) OPERATOR'S MANUAL B-82654EN/04 but changed for only R-30iB Plus/R-30iB Mate Plus.

In addition, following points are changed.

- When device is write protected, write is missed.
- Sometimes assignment clear procedure is necessary.
- Communication way over 16 bit integer or real number.
- Initial setting isn't necessary.



## 2 CC-Link INTERFACE (SLAVE) OVERVIEW

The CC-Link Interface (Slave) function on robot is a communication function that conforms to the protocol, CC-Link Version 1.10 of CLPA (CC-Link PARTNER ASSOCIATION). Robot with this function acts as a Remote Device station.

“CC-Link Remote Device Station PCB” hardware is necessary to use this function.

Cable length between stations should be 2m or more in CC-Link Ver.1.00 in specific condition, but cable length between stations should be 20cm or more in CC-Link Ver. 1.10. Wiring become easy in Ver.1.10. If Ver.1.10 device and Ver.1.00 device are mixed, total cable length and cable length between stations are Ver.1.00 specifications.

If system consists of only Ver.1.10 or later (Ver.1.10, Ver.2.00) devices and Ver.1.10 cables, cable length between stations is 20cm or more.

**System consists of only Ver.1.10 or later (Ver.1.10, Ver.2.00) devices and Ver.1.10 cables  
(Using End resistance110Ω)**

| Communication speed | Cable length between stations | Max total cable length |
|---------------------|-------------------------------|------------------------|
| 156kbps             | 20cm or more                  | 1200m                  |
| 625kbps             |                               | 900m                   |
| 2.5Mbps             |                               | 400m                   |
| 5Mbps               |                               | 160m                   |
| 10Mbps              |                               | 100m                   |

**Reference Only CC-Link Ver.1.00 case (Using End resistance110Ω)**

| Communication speed | Cable length between stations                                                 |                                                                       | Max total cable length |
|---------------------|-------------------------------------------------------------------------------|-----------------------------------------------------------------------|------------------------|
|                     | Between Remote I/O station or Remote device station<br>(Minimal-length cable) | Anteroposterior stations of Master / Local intelligent device station |                        |
| 156kbps             | 30cm or more                                                                  | 1m or more(A)<br>2m or more(B)                                        | 1200m                  |
| 625kbps             |                                                                               |                                                                       | 600m                   |
| 2.5Mbps             |                                                                               |                                                                       | 200m                   |
| 5Mbps               | 30cm~59cm *                                                                   |                                                                       | 110m                   |
|                     | 60cm or more*                                                                 |                                                                       | 150m                   |
| 10Mbps              | 30cm~59cm *                                                                   |                                                                       | 50m                    |
|                     | 60cm~99cm *                                                                   |                                                                       | 80m                    |
|                     | 1m or more                                                                    |                                                                       | 100m                   |

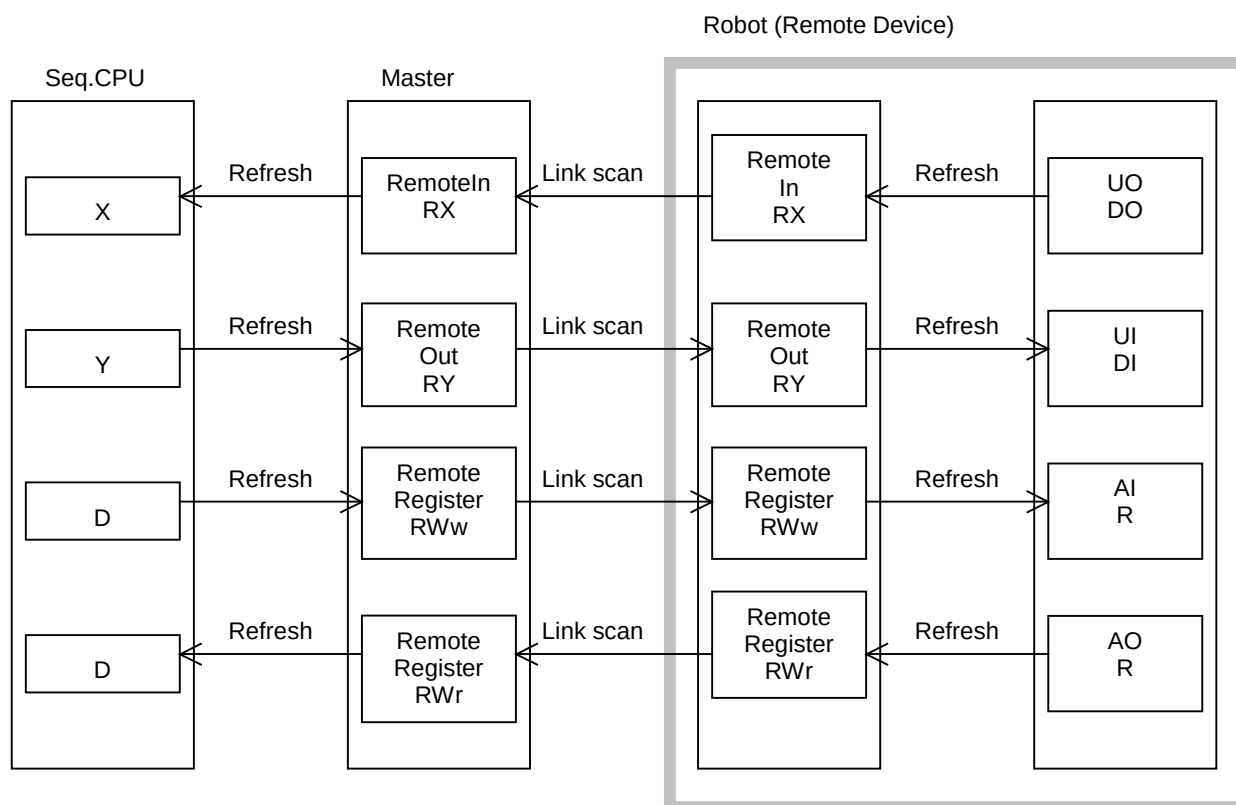
(A):1m or more is only when system consists of only master, remote I/O, remote device stations

(B):2m or more is when system also includes local stations or intelligent device stations

\* : If only 1 part of cable length between remote I/O or remote device stations is this condition, rightmost “Max total cable length” is adopted.

## 2.1 FEATURE

The CC-Link Interface (Slave) function exchanges Digital I/O data (DI/DO) and UOP data (UI/UO) via Remote Input/Output (RX/RX), and exchanges Analog I/O data (AI/AO) and Register data (R) via Remote Registers (RWr/RWw).



## 2.2 TERMS

| TERMS                                       | DESCRIPTION                                                                                                                                                                                                                   |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Remote Device station                       | A station that can exchange both bit data and 16 bit words.                                                                                                                                                                   |
| Remote Input :RX<br>/Remote Output:RY       | Bit data transmitted within a CC-Link network or an area stored with this bit data. Input data on the Master station is named RX and output data is named RY.                                                                 |
| Remote Register RWr<br>/Remote Register RWw | 16 bit words transmitted within a CC-Link network or an area stored with this word data. Input data on the Master station is named RWr and output data is named RWw.                                                          |
| CSP file                                    | CC-Link specification tries to load CSP file to master and master setting is done. But software which can load CSP file seems to be not exist. CSP file itself is prepared for FANUC. If you need CSP file, request to FANUC. |



## 2.3 SPECIFICATION OVERVIEW

Table 2.3 CC-Link interface (slave) specification overview

| ITEM                                                                                                        | SPECIFICATION                                                                                                                                                   |
|-------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Communication protocol                                                                                      | CC-Link Version 1.10                                                                                                                                            |
| Station type                                                                                                | Remote Device station                                                                                                                                           |
| Baud rate                                                                                                   | 165K, 625K, 2.5M, 5M or 10M bps can be selected.                                                                                                                |
| Number of CC-Link communication boards that can be installed on a robot.                                    | 2<br>Please see 1.4 ORDER NUMBER for the detail of CC-Link communication board.                                                                                 |
| Number of CC-Link Remote Device Stations by a CC-Link communication board.                                  | 1, 2, 3 or 4 stations can be occupied.                                                                                                                          |
| Number of data that can be exchanged by a CC-Link communication board.<br>(In case of 4 stations occupancy) | Remote In RX : User area 112 + System area 16*<br>Remote Out RY: User area 112 + System area 16*<br>Remote Register RWr: 16**<br>Remote Register RWw: 16**      |
| How to add the second CC-Link communication board.                                                          | Add another CC-Link Remote Device Station PCB to an empty slot of the controller.<br>(NOTICE:<br>Up to two CC-Link Remote Device Station PCB can be installed.) |

(\*) The usage of “System area” is defined in “CC-Link Specifications (Profile) BTP-05028” of CLPA. The “System area” can not be used for any purposes other than the defined usage.

(\*\*) The data is restricted to the patterns. Refer to the Section 2.3 Remote Register.

## 2.4 ORDER NUMBER

Table 2.4 (a) CC-Link interface (slave)

| Name                      | Order number |
|---------------------------|--------------|
| CC-Link Interface (Slave) | A05B-*-J786  |

“CC-Link Remote Device Station PCB” hardware is necessary to use this function.  
Up to 2 CC-Link Remote Device Station PCB can be installed to controller.

Table 2.4 (b) CC-Link remote device station PCB

| Name                              | Order number |
|-----------------------------------|--------------|
| CC-Link Remote Device Station PCB | A05B-*-J110  |



# 3 SETUP

This chapter describes the setting of CC-Link Interface (Slave) function.

## 3.1 CC-Link INTERFACE SETUP

The following table shows the setup items of CC-Link Interface (Slave) function. You must set up these items referring to their descriptions, before you use this function.

Table 3.1 Setup items of CC-Link interface

| ITEM                                                  | DESCRIPTION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Error one shot<br><br>default: DISABLE                | <p>ENABLE:<br/>The errors of CC-Link Interface (Slave) function can be reset, even while this function is in the error state. e.g. You should enable this item if you need to move the robot even while a data link error is occurring.</p> <p>DISABLE:<br/>The errors of CC-Link Interface (Slave) function can NOT be reset before their causes are resolved.</p> <p><b><u>NOTE) If this item is enabled, the robot can move even while this function is in the error state. Disable this item before starting a production.</u></b></p> |
| Station No.<br><br>range: 1 to 64<br>default: 1       | <p>This item indicates the station number. In case that two or more stations are occupied, enter the head number of these. For example,</p> <p>if StationNo.=10 and Number of Stations=4,<br/>then Station No.10, 11, 12 and 13 will be occupied.</p> <p>If Station No.65 or later is occupied, then "PRIO-322 St.No out of range" will be posted. Enter values that meet the following condition:<br/>StationNo. + Number of Stations - 1 ≤ 64</p>                                                                                        |
| Number of Stations<br><br>range: 1 to 4<br>default: 4 | <p>This item indicates the number of stations that will be occupied. The number of bit data and word data exchanged will be decided according to this item.</p> <p><b><u>NOTE) The last 16 points of bit data are assigned to DI/DO, but their usage is limited to the usage as system area. For example, user program can't turn on the DO in the system area.</u></b></p> <p><b><u>NOTE) The UOP(UI/UO) will NOT be assigned when the number of stations is 1.</u></b></p>                                                               |
| Baudrate<br><br>default: 10Mbps                       | <p>This item indicates the baud rate. The following baud rates are available.</p> <ul style="list-style-type: none"> <li>- 156Kbps</li> <li>- 625Kbps</li> <li>- 2.5Mbps</li> <li>- 5Mbps</li> <li>- 10Mbps</li> </ul>                                                                                                                                                                                                                                                                                                                     |
| RWr ( x )<br>(Not allowed to set)                     | <p>By the following setting, Analog output(AO) and Registers(R) will be assigned to Remote Registers(RWr). The number of RWr data will be decided according to the number of stations.</p> <p>(The number of RWr data 'x' will be shown in brackets).</p>                                                                                                                                                                                                                                                                                  |
| Number of AOs<br><br>range: 0 to 32766<br>default: 0  | <p>This item indicates the number of Analog outputs(AO) that will be assigned to RWr. Enter a value that meets the following condition:<br/>Number of AOs + Number of Registers ≤ x</p>                                                                                                                                                                                                                                                                                                                                                    |

| ITEM                                                            | DESCRIPTION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Number of Registers<br>range: 0 to 32766<br>default: 0          | This item indicates the number of Registers(R) that will be assigned to RWr.<br>Enter a value that meets the following condition:<br>Number of AOs + Number of Registers $\leq$ x                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Reg start index<br>range: 1 to 32766<br>default: 1              | This item indicates the head number of Registers(R) that will be assigned to RWr. For example,<br><br>if Number of Registers=2 and Reg start index=1,<br>then R[1] and R[2] will be assigned to RWr.<br><br><b><u>NOTE) Assign existent Registers.</u></b>                                                                                                                                                                                                                                                                                                                                                                                                     |
| RWw ( y )<br>(Not allowed to set)                               | By the following setting, Analog input(AI) and Registers(R) will be assigned to Remote Registers(RWw). The number of RWw data will be decided according to the number of stations.<br>(The number of RWw data 'y' will be shown in brackets).                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Number of AIs<br>range: 0 to 32766<br>default: 0                | This item indicates the number of Analog inputs(AI) that will be assigned to RWw. Enter a value that meets the following condition:<br>Number of AIs + Number of Registers $\leq$ y                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Number of Registers<br>range: 0 to 32766<br>default: 0          | This item indicates the number of Registers(R) that will be assigned to RWw.<br>Enter a value that meets the following condition:<br>Number of AIs + Number of Registers $\leq$ x                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Reg start index<br>range: 1 to 32766<br>default: 10             | This item indicates the head number of Registers(R) that will be assigned to RWw. For example,<br><br>if Number of Registers=5 and Reg start index=5,<br>then R[5], R[6], R[7], R[8] and R[9] will be assigned to RWw.<br><br><b><u>NOTE) Assign existent Registers.</u></b>                                                                                                                                                                                                                                                                                                                                                                                   |
| Reg Data<br>range: Unsigned Int or Int<br>default: Unsigned Int | This item specifies the data type of robot side receiving remote registers RWw that are assigned to Registers (R). You can select whether unsigned Integer (Range: 0 to 65535) or Integer (Range: -32768 to 32767).<br><br><b><u>NOTE)</u></b><br><b><u>1. This setting doesn't affect to robot side sending register RWr.</u></b><br><b><u>Accommodate the possible range of sending register (R[*]) and receiving side setting (Unsigned Int or Signed Int). If these aren't matched, invalid data exchange may happen.</u></b><br><b><u>2. Remote registers RWr or RWw assigned to analog input output AI/O are always handled as unsigned Integer.</u></b> |

CC-Link SETUP screen will be displayed by the following steps:

MENU → 6 SETUP → F1 TYPE → CC-Link

| SETUP CC-Link          |                | JOINT 10% |
|------------------------|----------------|-----------|
| Remote device board:1  |                | 1/10      |
| 1 Error one shot:      | DISABLE        |           |
| 2 Station No.:         | 1              |           |
| 3 Number of Stations:  | 4              |           |
| 4 Baudrate:            | [ 10Mbps ]     |           |
| RWr ( 16 )             |                |           |
| 5 Number of AOs:       | 0              |           |
| 6 Number of Registers: | 0              |           |
| 7 Reg start index:     | 1              |           |
| RWw ( 16 )             |                |           |
| 8 Number of AIs:       | 0              |           |
| 9 Number of Registers: | 0              |           |
| 10 Reg start index:    | 10             |           |
| 11 Reg Data:           | [Unsigned Int] |           |
| [ TYPE ]               | BOARD          | >         |
| CLA_ASG                |                | >         |

#### NOTE

- When you change "Number of Stations", "Number of AIs" or "Number of AOs", clear the current I/O assignment by pushing NEXT and F1(CLR\_ASG). The changes will be activated at the next power-on. Clear assignment may be necessary before changing assignment.  
If clear assignment isn't done, I/O communication may not be done. If it doesn't work fine, try clear assignment or do whenever possible.
- When you change "Station No." or its following items, cycle power to activate the changes.
- In case that two CC-Link communication boards are installed to a robot, the board which has a smaller physical slot number will be "Board1" and another will be "Board2". For information of the physical slot number, refer to "B.1 Installation".
- The values set on this screen will be saved in the system file "CCLINK.SV." To save settings of the CC-Link, press [FCTN] key and select "SAVE" when displaying any CC-Link screen. You can also save it from FILE screen -> F4 key BACKUP -> "1 System files" or "8 All of above". To select a device for saving, press [F5] "UTIL" on the file screen and select "Set Device".  
Please check whether CCLINK.SV is wrote because writing is missed when device is write protected even when prompt shows them as saved.
- If an out of range value is entered into a remote register, "PRIO-331 CC-LK Reg data invalid (Board No.)" will be posted and zero value will be sent to master device instead of the invalid register data.

## 3.2 REMOTE INPUT/OUTPUT( RX/RY )

The number of Remote input data(RX) and Remote output data(RY) exchanged by the CC-Link Interface function is decided according to the number of stations. The following table shows the relation between the number of RX/RY and the number of stations.

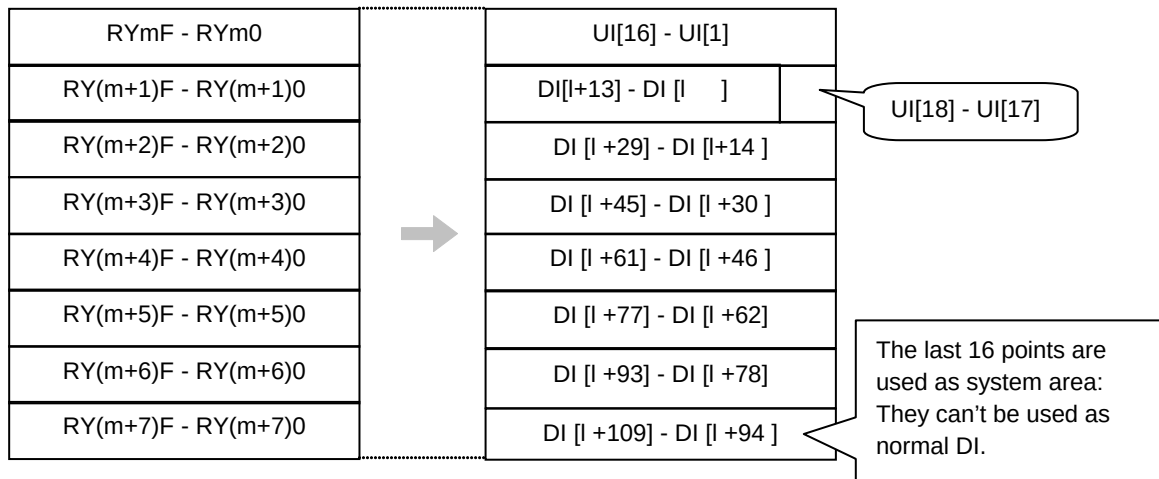
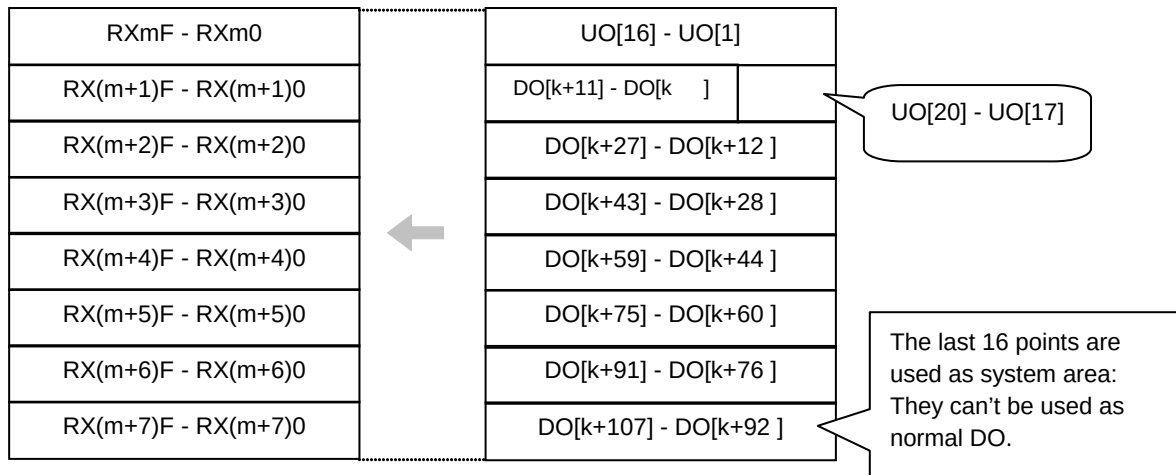
|                     | 1 station                                     | 2 stations                                    | 3 stations                                    | 4 stations                                     |
|---------------------|-----------------------------------------------|-----------------------------------------------|-----------------------------------------------|------------------------------------------------|
| Remote input<br>RX  | User area 16 pnts<br>+<br>System area 16 pnts | User area 48 pnts<br>+<br>System area 16 pnts | User area 80 pnts<br>+<br>System area 16 pnts | User area 112 pnts<br>+<br>System area 16 pnts |
| Remote output<br>RY | User area 16 pnts<br>+<br>System area 16 pnts | User area 48 pnts<br>+<br>System area 16 pnts | User area 80 pnts<br>+<br>System area 16 pnts | User area 112 pnts<br>+<br>System area 16 pnts |

UOP output (UO) and Digital output (DO) will be assigned to Remote input (RX) in turn. UOP input (UI) and Digital input (DI) will be assigned to Remote output (RY) in turn.

### NOTE

- 1 The last 16 points of DI/DO of CC-Link are used as system area, but their usage is limited to the usage as system area. For example, user program can't turn on the DO in the system area.
- 2 The following change of the system variable prevents the UOP from being assigned to RX/RY. When you change the system variable, clear the I/O assignment and cycle power.  
\$CCLINKRD[ board no. ].\$ASGUOP = TRUE -> FALSE
- 3 In case of 1 station, the UOP(UI/UO) will NOT be assigned.

e.g. Number of Stations=4, UO:20 points, UI:18 points.



#### NOTE

- 1 The value of 'm' will be decided according to the Station No. Refer to the CC-Link Master station manual for details.
- 2 The value of 'k' and 'l' will be decided by the system automatically but you can change them. To check and change the value, refer to the Digital I/O CONFIG screen. The following shows Rack No. and Slot No. of the CC-Link communication board.

|          |                          |
|----------|--------------------------|
| Rack No. | 92                       |
| Slot No. | Board 1: 1<br>Board 2: 2 |

- 3 The last 16 points of DI/DO of CC-Link are used as system area, and their usage is limited to the usage as system area. For example, user program can't turn on the DO in the system area.
- 4 The following change of the system variable prevents the UOP from being assigned to RX/RX. When you change the system variable, clear the I/O assignment and cycle power.  
\$CCLINKRD[ board no. ].\$ASGUOP = TRUE -> FALSE
- 5 In case of 1 station, the UOP(UI/UO) will NOT be assigned.

### 3.3 REMOTE REGISTER( RWr/RWw )

The number of Remote register data(RWr/RWw) exchanged by the CC-Link Interface function is decided according to the number of stations. The following table shows the relation between the number of RWr/RWw and the number of stations.

|                     | 1 station | 2 stations | 3 stations | 4 stations |
|---------------------|-----------|------------|------------|------------|
| Remote register RWr | 4 points  | 8 points   | 12 points  | 16 points  |
| Remote register RWw | 4 points  | 8 points   | 12 points  | 16 points  |

Analog output (AO) and Registers (R) will be assigned to Remote registers (RWr) in turn. Analog input (AI) and Registers (R) will be assigned to Remote registers (RWw) in turn.

The value exchanged via R is unsigned word data (16bits) of which the range is 0 to 65535, or signed word data (16bits) of which the range is -32768 to 32767. You can send float value, which is larger than -32769 and smaller than 65536, if \$CCLINKRD[board No.].\$AUTO\_REGCHG is set to not 0. The value is sent with integer by cutting off the floating part.

The value exchanged via AI/AO is always unsigned word data (16bits) of which the range is 0 to 65535 ( 0x0000 to 0xFFFF ).

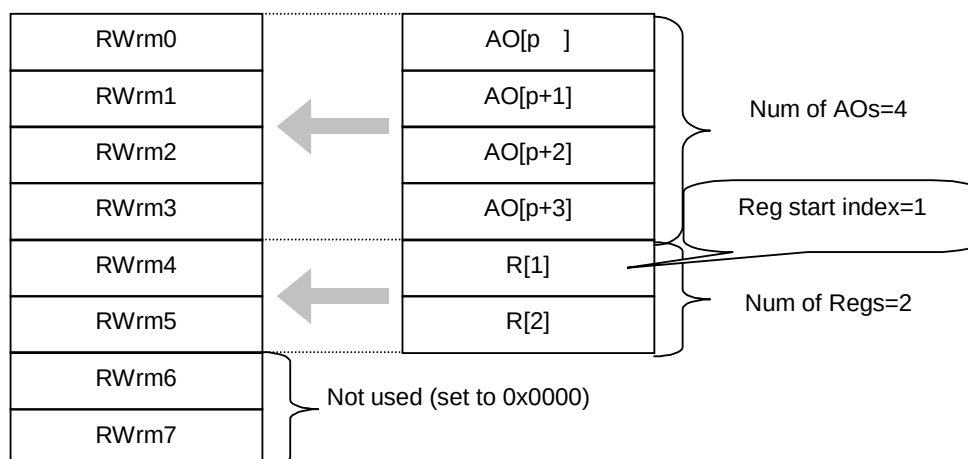
e.g. Assignment to RWr (2 stations)

Number of Stations = 2

Number of AOs = 4

Number of Registers = 2

Reg start index = 1



#### NOTE

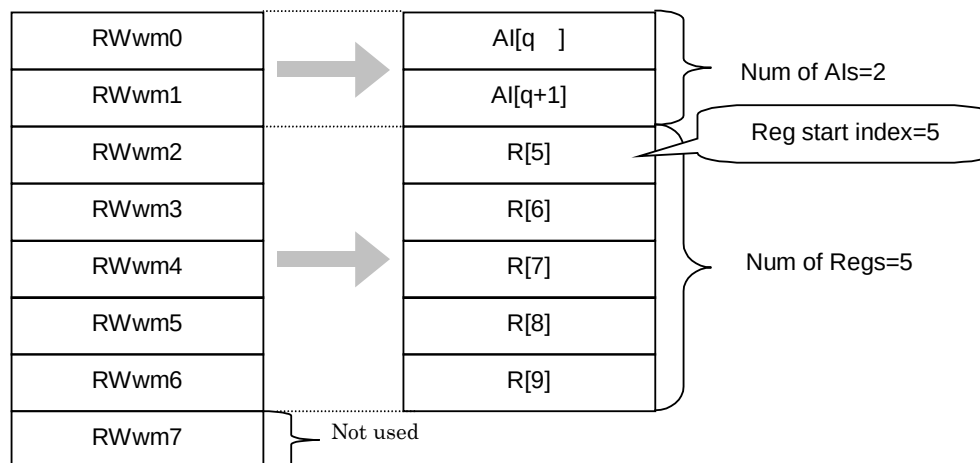
- 1 The value of 'm' will be decided according to the Station No. Refer to the CC-Link Master station manual for details
  - 2 The value of 'p' will be decided by the system automatically but you can change them. To check and change the value, refer to the Analog I/O CONFIG screen.
- The following shows Rack No. and Slot No. of the CC-Link communication board

|          |                          |
|----------|--------------------------|
| Rack No. | 92                       |
| Slot No. | Board 1: 1<br>Board 2: 2 |

e.g. Assignment to RWw (2 stations)



Number of Stations = 2  
 Number of AIs = 2  
 Number of Registers = 5  
 Reg start index = 5

**NOTE**

- 1 The value of 'm' will be decided according to the Station No. Refer to the CC-Link Master station manual for details
- 2 The value of 'q' will be decided by the system automatically but you can change them. To check and change the value, refer to the Analog I/O CONFIG screen. The following shows Rack No. and Slot No. of the CC-Link communication board.

|          |                          |
|----------|--------------------------|
| Rack No. | 92                       |
| Slot No. | Board 1: 1<br>Board 2: 2 |

### 3.3.1 How to send the integer other than 16 bit (signed / unsigned) integer or real number

You should use 2 register to send them. Concretely, for example, set integral part of the value, which is divided by 65536 from sent value, to R[2], set the odd to R[1], and send both register by CC-Link.

Example of TP program (When you want to send R[100])

R[1] = R[100] MOD 65536

// Pay attention R[100] isn't changed at here.

R[2] = R[100] DIV 65536

When you want to receive the value larger than 65535, convert the value received by 2 register.

Example of TP program (When you convert received data by R[1] and R[2] to R[100])

R[100] = R[2] \* 65536 + R[1] (Pay attention R[1] and R[2] are changed during running this line.)

You can send the value less than signed 16 bit value or real number by similar way.



# 4 STATUS

The current status of the Remote Device station is indicated by the LEDs placed on the CC-Link communication board and displayed on the CC-Link STATUS screen.

## 4.1 LED

The CC-Link Remote Device Station communication board has 4 LEDs. The following table shows their functions.

| NAME | COLOR | ON                                                   | OFF                                                                                                                                |
|------|-------|------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| ERR  | Red   | - CRC error occurred<br>- Station No. setup abnormal | - Normal data exchange<br>- Hardware being reset                                                                                   |
| RD   | Green | Data being received                                  | - Failure in receiving data<br>- Hardware being reset                                                                              |
| SD   | Green | Data being sent                                      | - No data sent<br>- Hardware being reset                                                                                           |
| RUN  | Green | In the data link                                     | - Before entering the data link<br>- Failure in receiving data<br>- Time out occurred in receiving data.<br>- Hardware being reset |

### NOTE

In case that ERR is OFF but the robot can't communicate with PLC even if the setting of the robot is correct, the station type of the robot in the setting of **PLC** might be "Remote I/O Station". Set the station type of the robot to "Remote Device Station" because robot only supports "Remote Device Station".

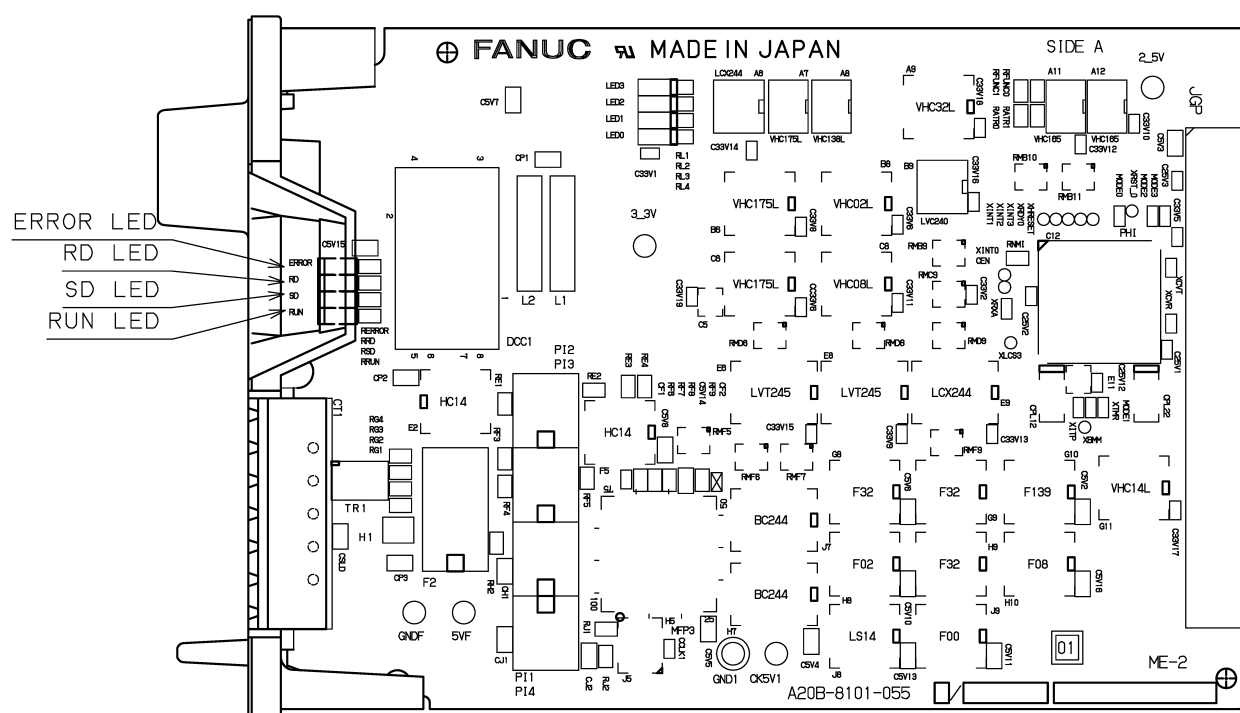


Fig. 4.1 Remote device station PCB

## 4.2 CC-LINK INTERFACE STATUS

The following table shows the items of the CC-Link STATUS screen.

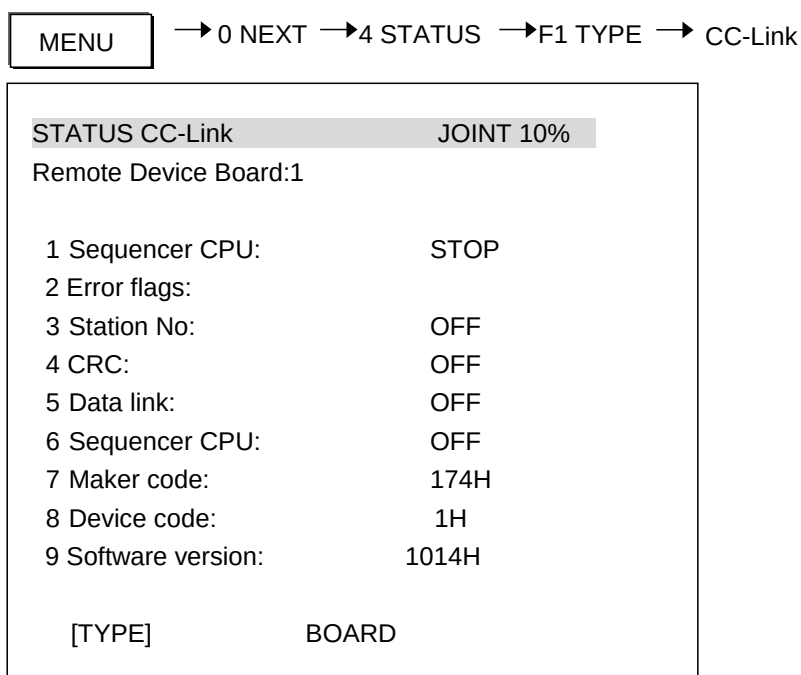
Table 4.2 Status items of CC-Link interface

| ITEM                        | DESCRIPTION                                                                                                                                                                                                       |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Sequencer CPU<br>STOP / RUN | This item indicates the status of the sequencer CPU* on the Master station. <b><u>This item indicates the correct status only while the Remote Device station is exchanging data with the Master station.</u></b> |
| Error flags<br>OFF / ON     | The following error flags indicate whether an error occurred or not for their factor.<br>( OFF: No error occurred. / ON: Error occurred )                                                                         |
| Station No                  | If ON, this Remote Device station occupied a station number that is out of the range (1 to 64).                                                                                                                   |
| CRC                         | If ON, CRC error occurred.                                                                                                                                                                                        |
| Data link                   | If ON, this Remote Device Station is out of the data link. <b><u>If you turn on the Remote Device station while the CC-Link cable is disconnected or the Master station is off, this item will NOT be ON.</u></b> |
| Sequencer CPU               | If ON, the sequencer CPU* on the Master station is abnormal. <b><u>This item indicates the correct status only while the Remote Device station is exchanging data with the Master station.</u></b>                |
| Maker code                  | This item indicates the maker code. **The maker code is "316H".                                                                                                                                                   |
| Device code                 | This item indicates the device code. **The device code is "23H".                                                                                                                                                  |
| Software version            | This item indicates the software version.                                                                                                                                                                         |

(\*) The sequencer CPU controls the units within the CC-Link network by executing a sequence program.

(\*\*) Maker code and Device code are defined by CLPA.

**CC-Link STATUS screen will be displayed by the following steps:**



# 5

## ALARM CODES

---

**PRIO-320 STOP CC-LK System error (ID=x)(y)**

**Cause** :Internal system error occurred.

**Remedy** :Contact FANUC's service site. The ID number in the brackets is necessary to track the problems.

**PRIO-321 WARN CC-LK Board not installed**

**Cause** :No CC-Link Remote Device Station PCB is installed.

**Remedy** :Install CC-Link Remote Device Station PCB.

**PRIO-322 STOP CC-LK St.No out of range (Board No.)**

**Cause** :A station number out of the range (1 to 64) was occupied.

**Remedy** :Change Station No. or Number of Stations on CC-Link SETUP screen.

**PRIO-323 WARN CC-LK CRC error (Board No.)**

**Cause** :CRC error was detected.

**Remedy** :Confirm the connection of CC-Link cable and terminal registers, and countermeasures to noise.  
Set Baud rate to match the Master station's baud rate.

**PRIO-324 STOP CC-LK Data link error (Board No.)**

**Cause** :This Remote Device station left the data link. The CC-Link cable were disconnected or the Master station was turned off.

**Remedy** :Confirm the connection of CC-Link cable and the status of the Master station.

**⚠ CAUTION**

If you turn on the Remote Device station while the CC-Link cable is disconnected or the Master station is off, this alarm will NOT be posted.

**PRIO-325 STOP CC-LK Seq. CPU stopped (Board No.)**

**Cause** :The sequencer CPU stopped.

**Remedy** :Confirm the status of the sequencer.

**⚠ CAUTION**

This alarm is detected correctly only while the Remote Device station is exchanging data with the Master station.

**PRIO-326 STOP CC-LK Seq. CPU abnormal (Board No.)**

**Cause** :The sequencer CPU is abnormal.

**Remedy** :Confirm the status of the sequencer.

**⚠ CAUTION**

This alarm is detected correctly only while the Remote Device station is exchanging data with the Master station.

**PRIO-327 STOP CC-LK Reg index error (Board No.)**

**Cause** :A non-existent register was used.

**Remedy** :Change Number of Registers or Reg start index on CC-Link SETUP screen.

**PRIO-328 STOP CC-LK AO/R too many (Board No.)**

**Cause:** Too many AOs or Registers were assigned to RWr.

**Remedy:** Change Number of AOs and Number of Registers on CC-Link SETUP screen.

**PRIO-329 STOP CC-LK AI/R too many (Board No.)**

**Cause:** Too many AIs and Registers were assigned to RWw.

**Remedy:** Change Number of AIs and Number of Registers on CC-Link SETUP screen.

**PRIO-330 STOP CC-LK PNTtoUOP not enough (Board No.)**

**Cause:** The number of points of RX/Ry to be assigned to UOP is not enough.

**Remedy:** Change Number of Stations on CC-Link SETUP screen.

**PRIO-331 STOP CC-LK Reg data invalid (Board No.)**

**Cause:** A numerical value set to a remote register was not 16bits integer.

**Remedy:** CC-Link function of robot supports only unsigned or signed 16bits word data. Do not enter another type of value to remote register.

You can send float value, which is larger than -32769 and smaller than 65536, if \$CCLINKRD[board No.].\$AUTO\_REGCHG is set to not 0. The value is sent with integer by cutting off the floating part.

Severity of this alarm can be changed to warning if \$CCLINKRD[board No.].\$REG\_ER\_WARN is set to not 0.

**PRIO-332 STOP CC-LK no comm. to master (Board No.)**

**Cause:** Channel carrier detection error.

**Remedy:** Confirm the connection of CC-Link cable and terminal registers, and countermeasures to noise. Set Baud rate to match the Master station's baud rate.

**PRIO-333 SYST CC-LK comm. task failed.**

**Cause:** Internal system error occurred.

**Remedy:** Contact FANUC's service site.

**PRIO-334 STOP CC-LK too many boards found**

**Cause:** Three or more CC-Link Remote Device Station PCB are installed.

**Remedy:** Robot controller supports only two CC-Link communication boards.  
Please turn off the controller and remove excess board.

# APPENDIX





# A MEMORY MAPPED PROFILE

The usage of Remote Input/Output(RX/RX) and Remote registers(RW<sub>r</sub>/RW<sub>w</sub>) is defined in “CC-Link Specifications (Profile) BTP-05028” of CLPA.

## A.1 REMOTE INPUT RX/REMOTE OUTPUT RY

The following shows the memory mapped profile of Remote Input(RX) and Remote Output(RY). The set of 16 bits located at the bottom is “System area.” The “System area” can not be used for any purposes other than the usage defined by CLPA.

Initial data setting isn't necessary for CC-Link of robot controller.

| Remote Device -> Master |                                      | Master -> Remote Device |                                       |
|-------------------------|--------------------------------------|-------------------------|---------------------------------------|
| Device No.              | Name                                 | Device No.              | Name                                  |
| RXm0                    | User area                            | RYm0                    | User area                             |
| RXm1                    |                                      | RYm1                    |                                       |
| RXm2                    |                                      | RYm2                    |                                       |
| RXm3                    |                                      | RYm3                    |                                       |
| RXm4                    |                                      | RYm4                    |                                       |
| RXm5                    |                                      | RYm5                    |                                       |
| RXm6                    |                                      | RYm6                    |                                       |
| RXm7                    |                                      | RYm7                    |                                       |
| RXm8                    |                                      | RYm8                    |                                       |
| RXm9                    |                                      | RYm9                    |                                       |
| RXmA                    |                                      | RYmA                    |                                       |
| RXmB                    |                                      | RYmB                    |                                       |
| RXmC                    |                                      | RYmC                    |                                       |
| RXmD                    |                                      | RYmD                    |                                       |
| RXmE                    |                                      | RYmE                    |                                       |
| RXmF                    |                                      | RYmF                    |                                       |
| ---                     |                                      | ---                     |                                       |
| RX(m+n)0                | Reserved                             | RY (m+n)0               | Reserved                              |
| RX(m+n)1                |                                      | RY (m+n)1               |                                       |
| RX(m+n)2                |                                      | RY (m+n)2               |                                       |
| RX(m+n)3                |                                      | RY (m+n)3               |                                       |
| RX(m+n)4                |                                      | RY (m+n)4               |                                       |
| RX(m+n)5                |                                      | RY (m+n)5               |                                       |
| RX(m+n)6                |                                      | RY (m+n)6               |                                       |
| RX(m+n)7                |                                      | RY (m+n)7               |                                       |
| RX(m+n)8                | Initial data processing request flag | RY (m+n)8               | Initial data processing complete flag |
| RX(m+n)9                | Initial data setting complete flag   | RY (m+n)9               | Initial data setting request flag     |
| RX(m+n)A                | Error status flag                    | RY (m+n)A               | Error reset request flag              |
| RX(m+n)B                | Remote READY                         | RY (m+n)B               | Reserved                              |
| RX(m+n)C                | Reserved                             | RY (m+n)C               |                                       |
| RX(m+n)D                |                                      | RY (m+n)D               |                                       |
| RX(m+n)E                |                                      | RY (m+n)E               |                                       |
| RX(m+n)F                |                                      | RY (m+n)F               |                                       |

m: This value is decided according to the station number.

n: This value is decided according to the number of stations.

| Device No. | Name                                  | Function                                                                                                                                                                                          |
|------------|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RX(m+n)8   | Initial data processing request flag  | This signal will be ON after power-on to request the Master station to perform the initial data processing. When the initial data processing is completed (RY(m+n)8=ON), this signal will be OFF. |
| RX(m+n)9   | Initial data setting complete flag    | In the Remote READY status, when the Master station requests the initial data setting (RY(m+n)9=ON), this signal will be ON.<br>When RY(m+n)9 is OFF, this signal will be OFF.                    |
| RX(m+n)A   | Error status flag                     | In the Remote READY status, when an error of this CC-Link Interface function occurs, this signal will be ON.<br>NOTE ) This signal reflects the errors of the CC-Link Interface function only.    |
| RX(m+n)B   | Remote READY                          | After power-on, when the initial data processing on the Master station is completed (RY(m+n)8=ON), this signal will be ON.                                                                        |
| RY(m+n)8   | Initial data processing complete flag | After power-on, when the initial data processing on the Master station is completed, this signal will be ON.                                                                                      |
| RY(m+n)9   | Initial data setting request flag     | When the Master station requests the initial data setting, this signal will be ON.                                                                                                                |
| RY(m+n)A   | Error reset request flag              | When this signal is turned ON, Error status flag(RX(m+n)A) will be OFF.<br>NOTE) Even when this signal is turned ON, the robot will NOT be reset.                                                 |

## A.2 REMOTE REGISTER RW<sub>r</sub> / RW<sub>w</sub>

The following shows the memory mapped profile of Remote Registers(RW<sub>r</sub>/RW<sub>w</sub>).

| Remote Device -> Master |            | Master -> Remote Device |            |
|-------------------------|------------|-------------------------|------------|
| Device No.              | Name       | Device No.              | Name       |
| RW <sub>rm</sub> 0      | User area  | RW <sub>wm</sub> 0      | User area  |
| RW <sub>rm</sub> 1      |            | RW <sub>wm</sub> 1      |            |
| RW <sub>rm</sub> 2      |            | RW <sub>wm</sub> 2      |            |
| RW <sub>rm</sub> 3      |            | RW <sub>wm</sub> 3      |            |
| RW <sub>rm</sub> 4      | 1 station  | RW <sub>wm</sub> 4      | 1 station  |
| RW <sub>rm</sub> 5      |            | RW <sub>wm</sub> 5      |            |
| RW <sub>rm</sub> 6      |            | RW <sub>wm</sub> 6      |            |
| RW <sub>rm</sub> 7      |            | RW <sub>wm</sub> 7      |            |
| RW <sub>rm</sub> 8      | 2 stations | RW <sub>wm</sub> 8      | 2 stations |
| RW <sub>rm</sub> 9      |            | RW <sub>wm</sub> 9      |            |
| RW <sub>rm</sub> A      |            | RW <sub>wm</sub> A      |            |
| RW <sub>rm</sub> B      |            | RW <sub>wm</sub> B      |            |
| RW <sub>rm</sub> C      | 3 stations | RW <sub>wm</sub> C      | 3 stations |
| RW <sub>rm</sub> D      |            | RW <sub>wm</sub> D      |            |
| RW <sub>rm</sub> E      |            | RW <sub>wm</sub> E      |            |
| RW <sub>rm</sub> F      |            | RW <sub>wm</sub> F      |            |
|                         | 4 stations |                         | 4 stations |

m: This value is decided according to the station number.

# B

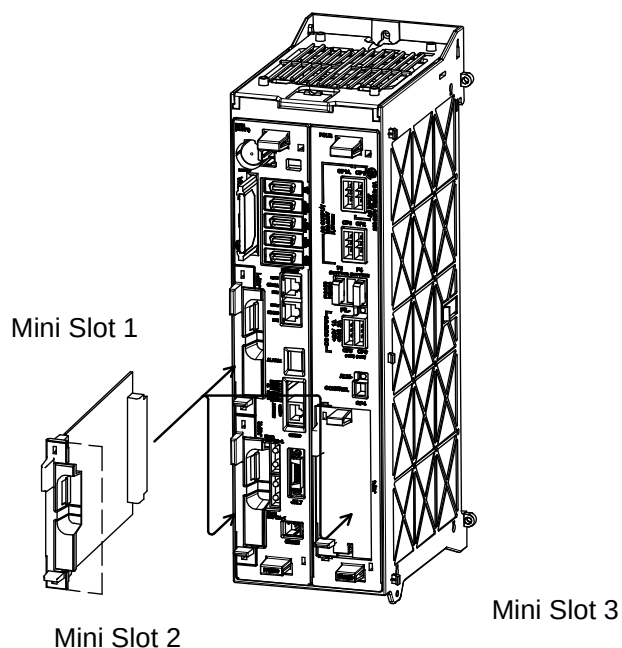
## CC-Link HARDWARE CONNECTION AND MAINTENANCE

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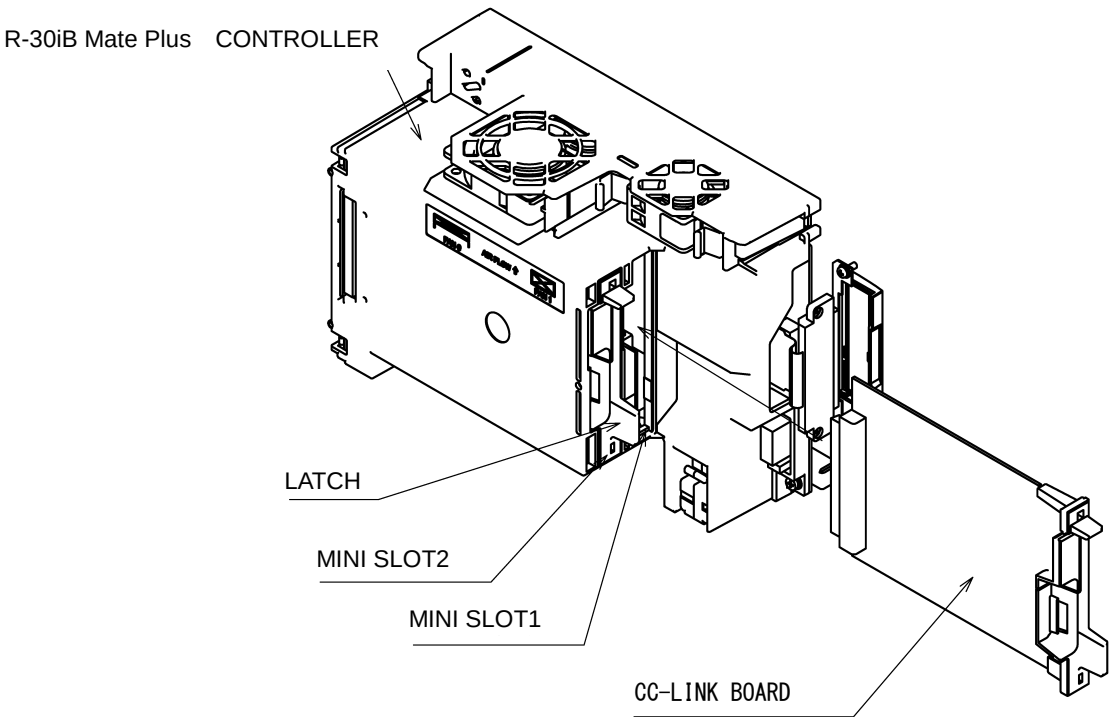
### B.1 INSTALLATION OF CC-Link REMOTE DEVICE STATION PCB

---

In the R-30/B Plus controller, a CC-Link Remote Device Station PCB is installed to an option slot like the figure below.



In the R-30iB Mate Plus controller, a CC-Link Remote Device Station PCB is installed to an option slot like the figure below.

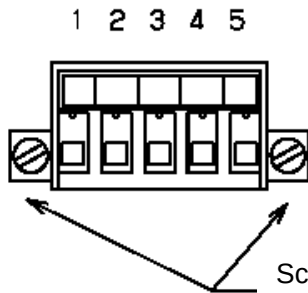


## B.2 CONNECTOR

The following connector is used as a CC-Link interface.

CT 1

Connector pin's array



| No. | Signal |
|-----|--------|
| 1   | DA     |
| 2   | DB     |
| 3   | DG     |
| 4   | SLD    |
| 5   | FG     |

Connector on the cable side:

Phoenix Contact

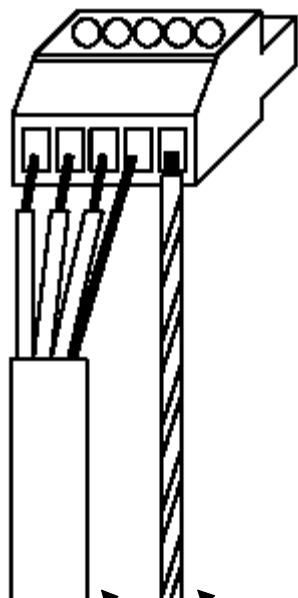
MSTB2.5/5-STF-5.08

(Attached with the product)

Applicable wires: AWG24-12

## B.3 HOW TO CONNECT A CC-Link CABLE

Loosen the screws to detach the connector from the PCB. Connect the communication cable to the connector by the following steps:



- (1) Pare the sheath of the cable and remove an unnecessary portion of the shield.
- (2) Pare the sheath of the wires to match the length of the terminals.(NOTE)
- (3) Loosen the screws to facilitate inserting wires.
- (4) Insert each wires into the holes of the connector.
- (5) Tighten the screws to fix the wires.
- (6) Attatch the connector to PCB.
- (7) Tighten the screws to fix the connector.

### NOTE

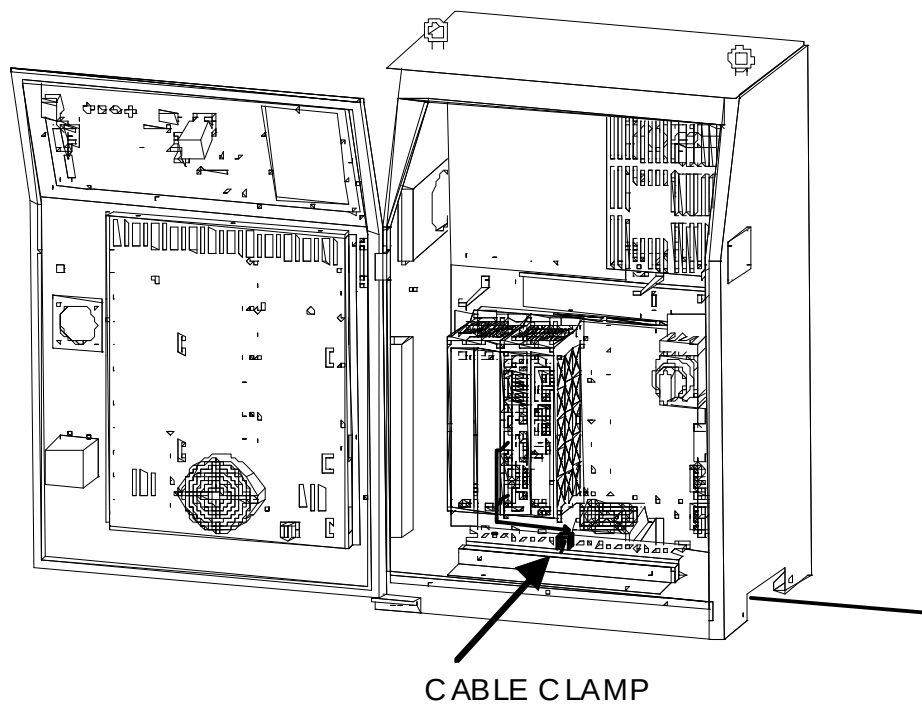
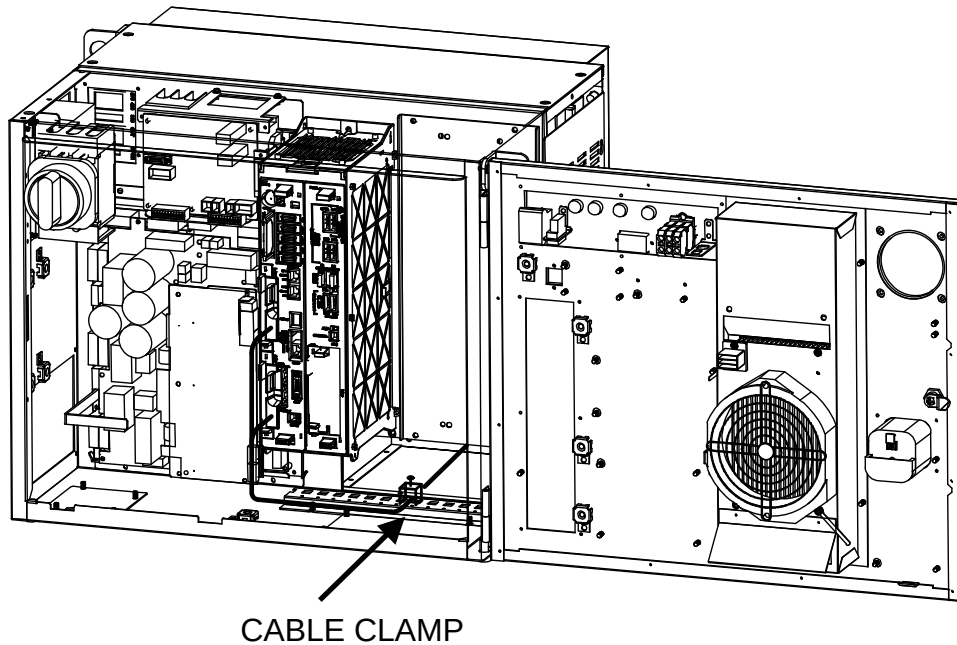
Terminals are available. Tie up the stripped wires and fix a terminal to it. The following terminal is recommended.  
Phoenix Contact, series AI (Applicable tool:ZA3)

FG cable are connected to both a connector and a cabinet before the delivery  
(Except the case of ordering PCB only).

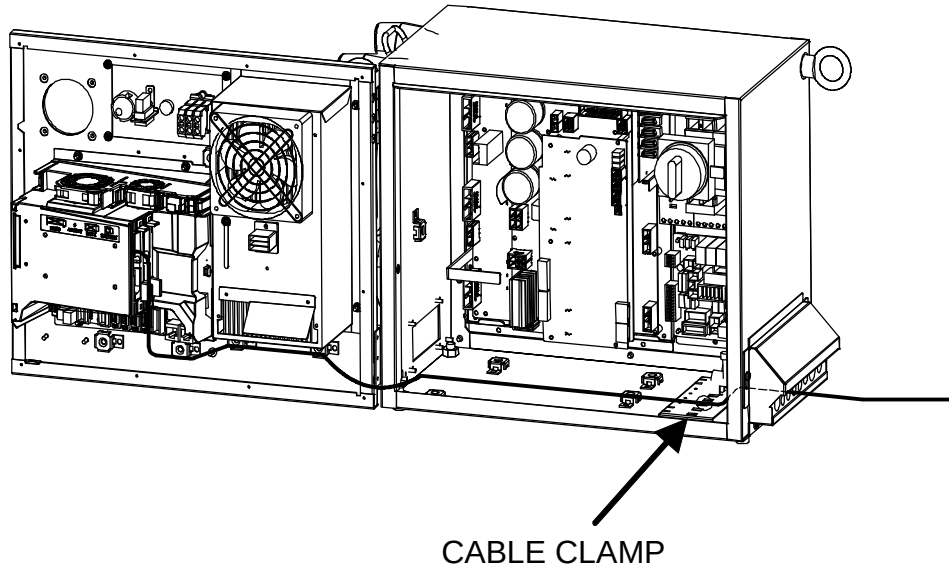
CC-Link cable needs to be prepared and connected by the user.

## B.4 HOW TO LEAD CC-Link CABLE

The following is how to lead the cable into the R-30iB Plus controller.

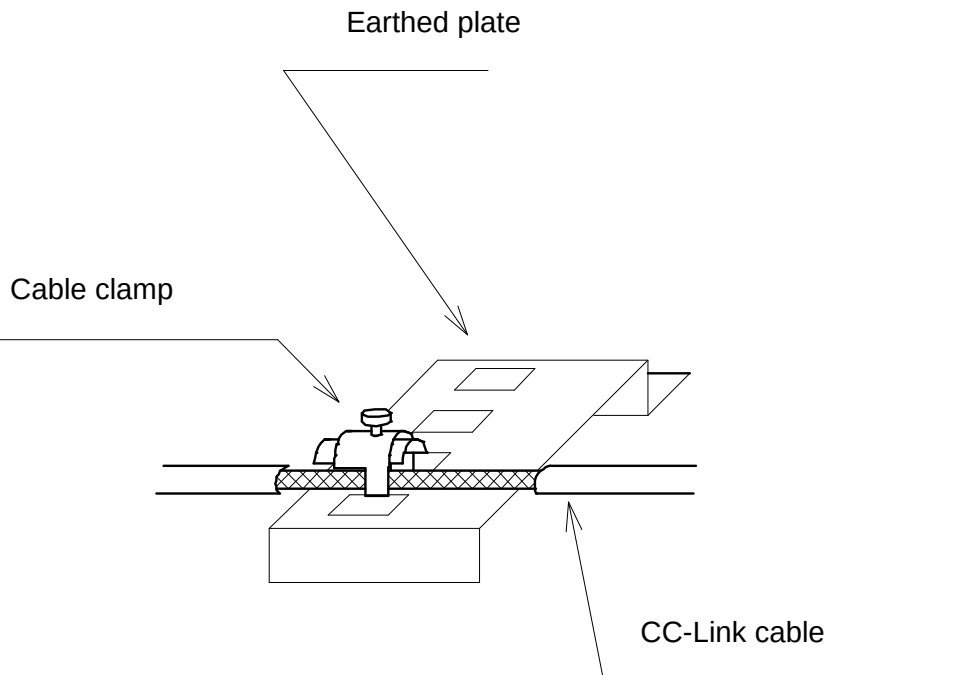


The following is how to lead the cable into the R-30iB Mate Plus controller.



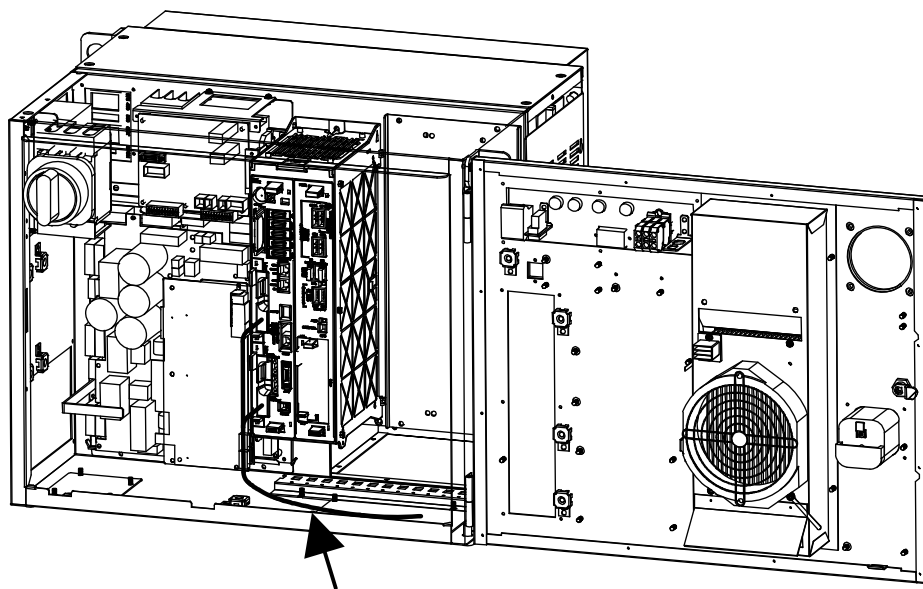
## B.5 HOW TO WIRE CC-Link CABLE

For information on how to wire the CC-Link cable, refer to the “CC-Link cable wiring manual” of CLPA. You must prepare the CC-Link cable and the terminal resisters. When you wire the CC-Link cable in the robot controller, you must pare the sheath of the CC-Link cable to expose the outer shield and fix the cable on the earthed plate by a clamp as follows. (For this purpose, a clamp is provided with a controller). Since this cable clamp is not only to fix the cable on the controller but also to stabilize the system, these steps must be done.

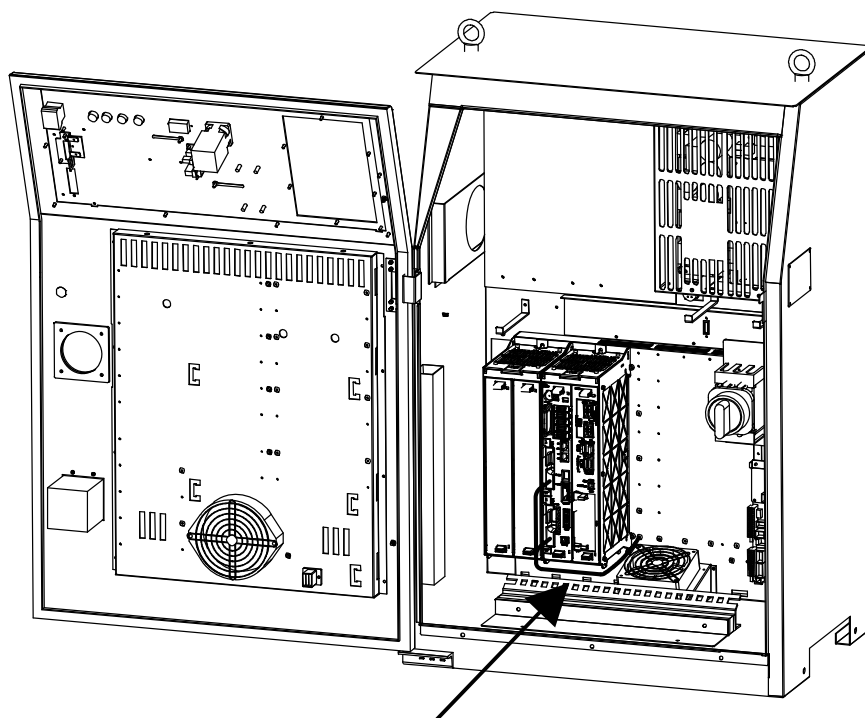


## B.6 HOW TO CONNECT THE EARTH CABLE IN CASE ONLY THE PCB IS PROVIDED

For R-30iB Plus controller, connect the bar terminal of the earth cable to the connector like “B.3 How to connect a CC-Link cable” and then connect the circle terminal to the robot cabinet to earth as follows.



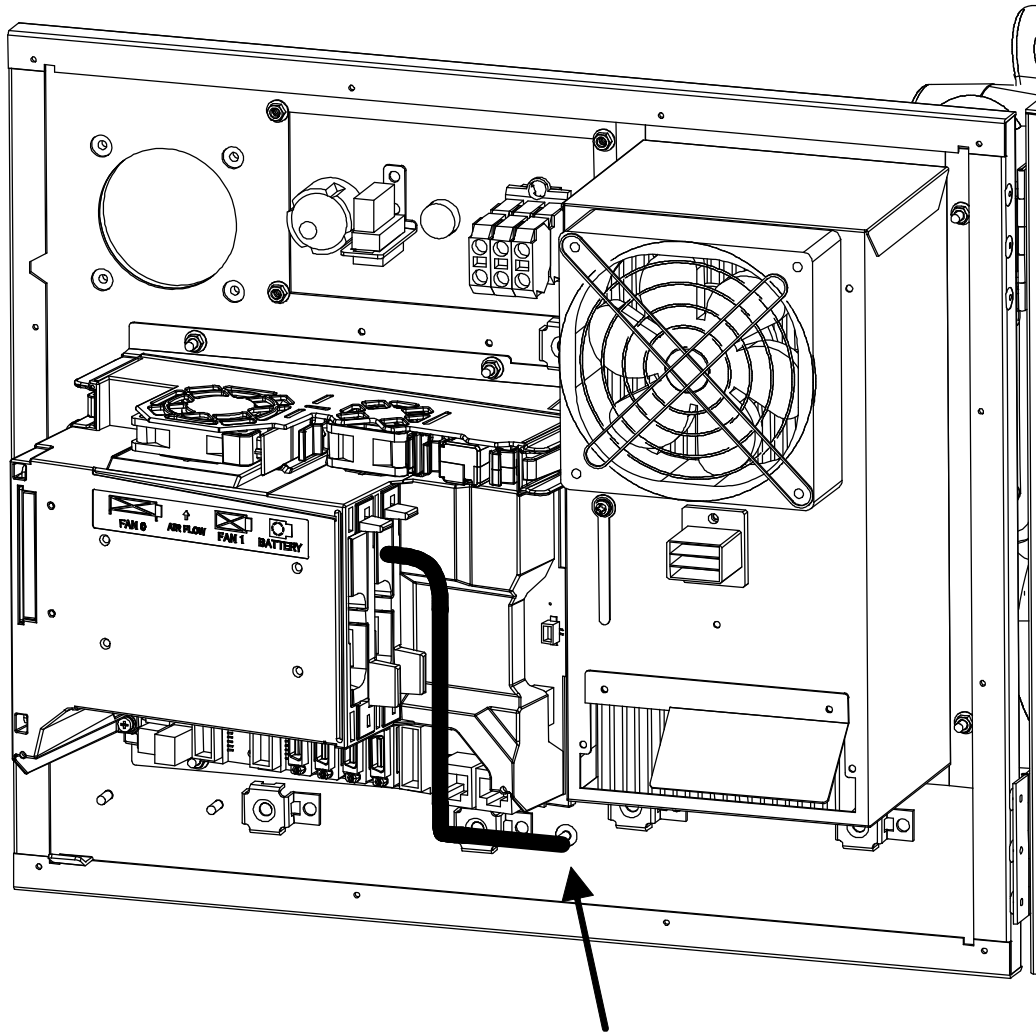
EARTH CABLE (M5 NUT)



EARTH CABLE (M5 SCREW)



For R-30iB Mate Plus controller, connect the circle terminal to the robot cabinet to earth as follows.



EARTH CABLE (M5 NUT)



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# REVISION RECORD

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**B-83974EN/01**

